

Small theories of large models

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E.T.: Empirical Theory in Representation Learning, ECCV 2026, Malmö, 8 September

The trouble about progress is that it always looks much greater than it really is.

Johann Nestroy.

What would it mean to have a theory of representation learning?

And who would hold it?

Theories, what are they good for

Two paradigms of perception

Mind-sized

Two levels

The approach

The example

Empirical theory

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Empirical theory

Understanding a system: three things you can do

Predict

Say what it will do.

Criterion: held-out data

Control

Know what to change.

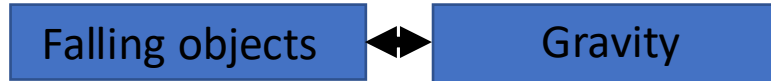
Criterion: held-out interventions

Extend

Take it to a new situation.

Criterion: a situation it was not made for
(build a bridge)

Worth something only out of distribution. The further out, the more it is worth.



The scientific method

Galileo → innovative combination of experimentation and mathematical modelling resulting in the start of the ‘modern’ scientific method.



Galileo

What is a good scientific explanation (covering law model)

(C) Condition/Fact	← <i>Explanans</i>	Simon Stevin drops a ball from the church tower of delft (108.75 m). $x = at^2$, $a = x/t^2$, $a = 9.81$, $t = \sqrt{x/a}$
(G) General Law		
(D) Descriptive Statement	← <i>Explanandum</i>	The ball drops in 3.3 seconds.

Hempel,
1952

For this situation:

Understanding == Predicting

Simple formulas can describe (even) the most fundamental phenomena

Scientific method

In this example, the three are one

Predict

the time of the drop

Control

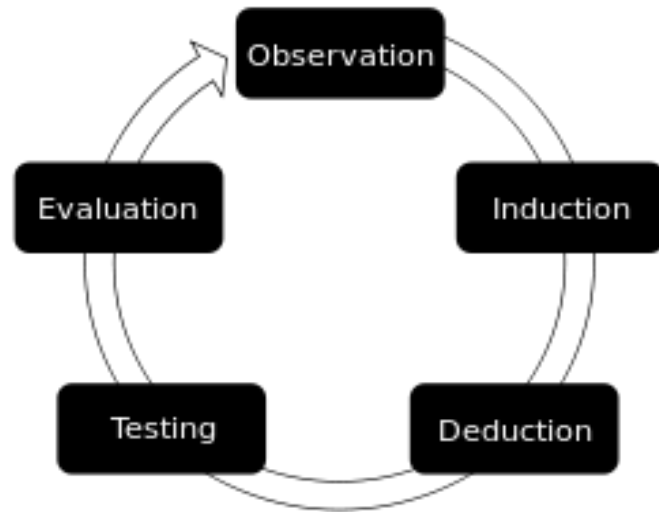
Feather falls slower than a hammer. But not if you pump out air.

Extend

1971 (apollo moon mission), a feather and a hammer land at the same time at the moon.

Understanding = predicting = control = extend.

Bridging the behavioural gap



A.D. de Groot, 1958

In the Empirical cycle you develop a **nomological network** from which a **hypothesis** is derived.

Causal test that **isolates one factor**

Believe that if we discover the correct causal chains and/or laws these can be subsequently used for prediction.

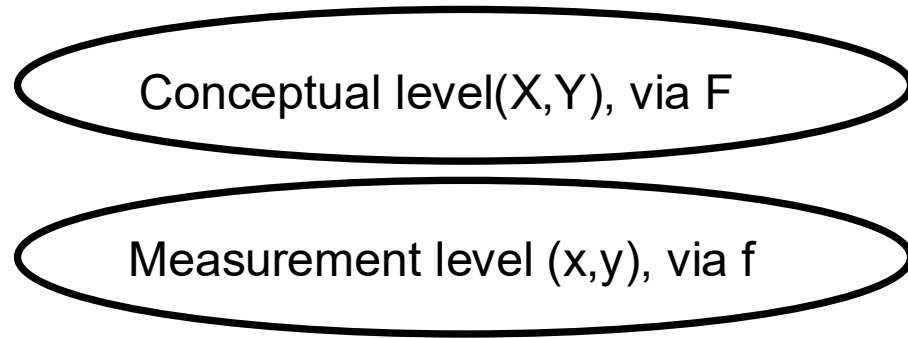
The ideal: a theory that, by understanding it, also results in prediction, control and the possibility to extend.

For this to work it implies:

- few laws.
- that are stable.
- Therefore small enough for a person to hold in mind (mind sized).

Modern machine learning starts with predicting and not theory

Explaining and predicting are not the same goals



Explain

estimate F by having f match F
 F is the (latent) function.
 f is used to test hypothesis (in hindsight).

Predict

estimate f on the basis of x and y .
 f represents the association between X and Y .
 f is used to predict future values of y on the basis of x

Theories, what are they good for

Two paradigms of perception

Mind-sized

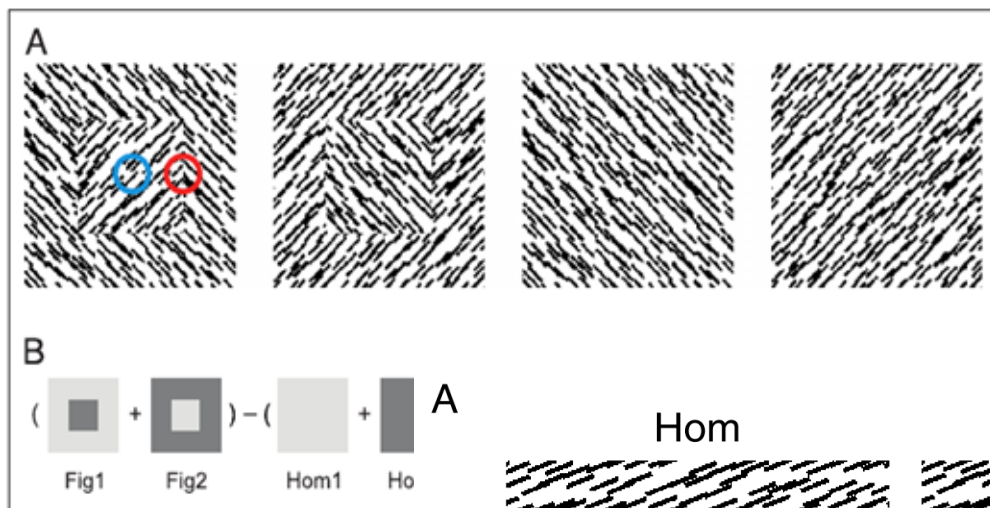
Two levels

The approach

The example

Empirical theory

SCENE SEGMENTATION

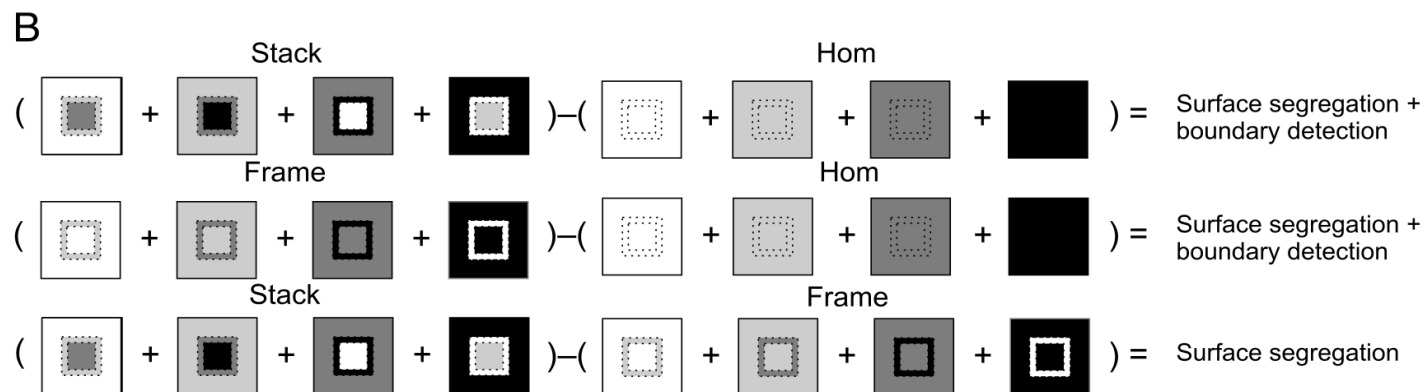
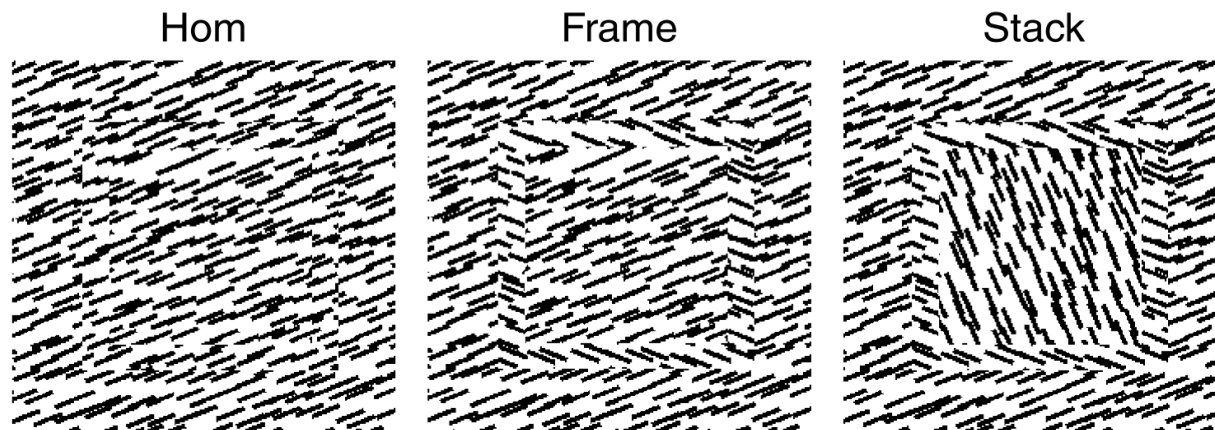


During my PhD

Carefull controlled stimuli with 1 clear segmentation source

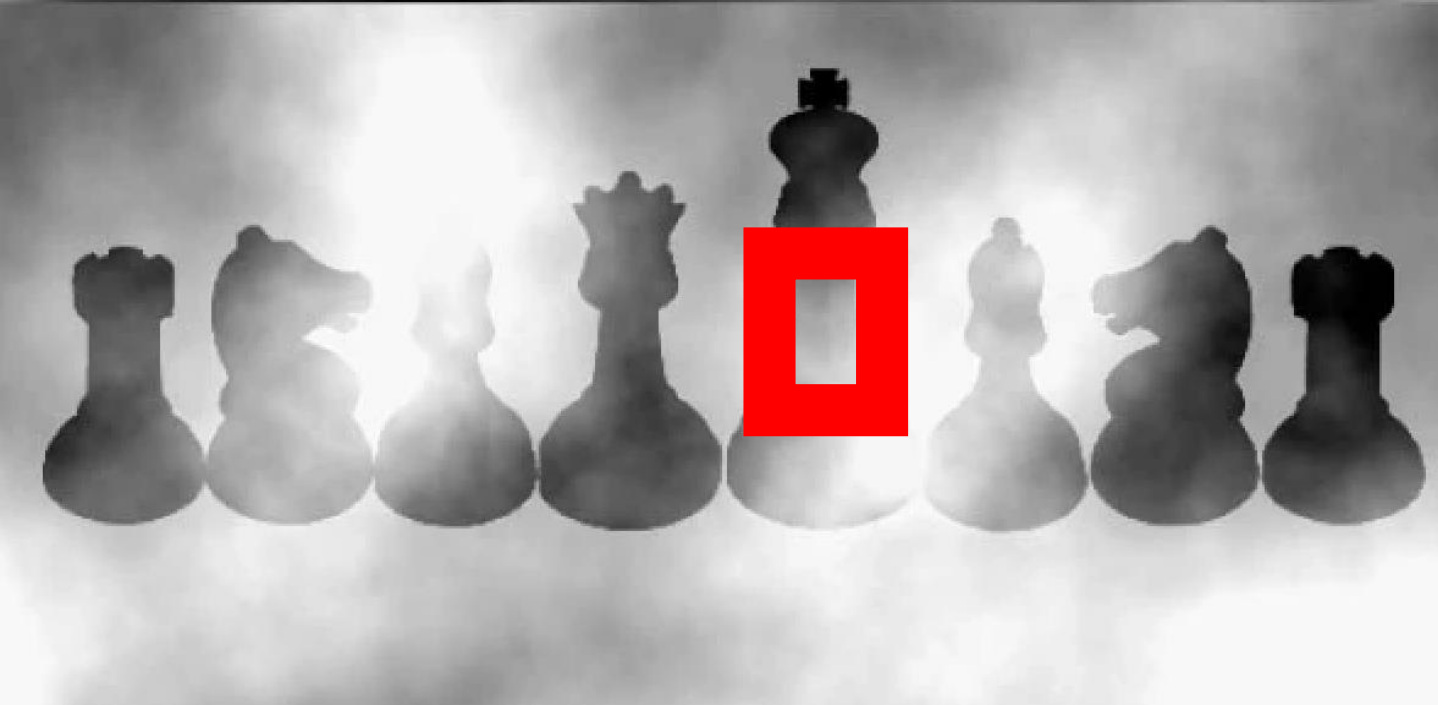
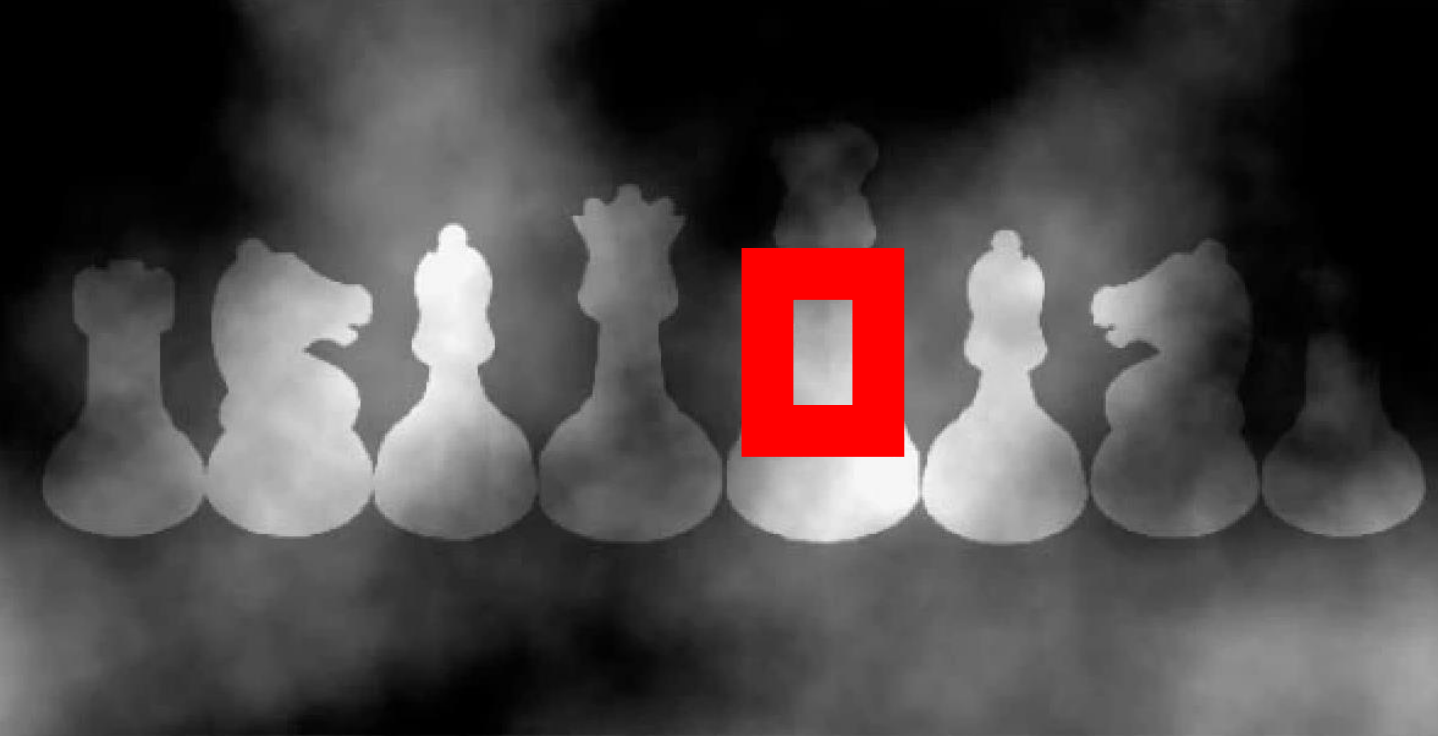
Avoid multi-cue , and 'easy' examples → Looking for the scene segmentation operation.

This way of thinking is typical for reconstruction theories of perception.

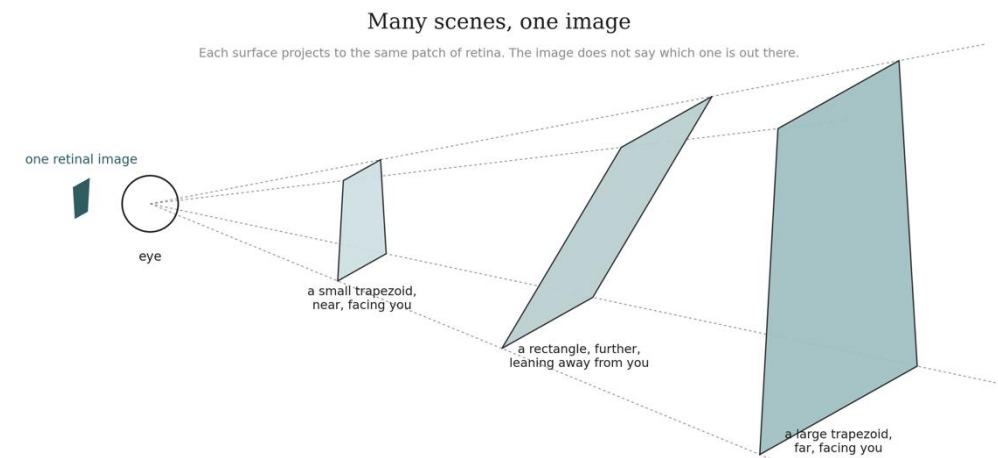


Scholte et al., 2008.

JoCN

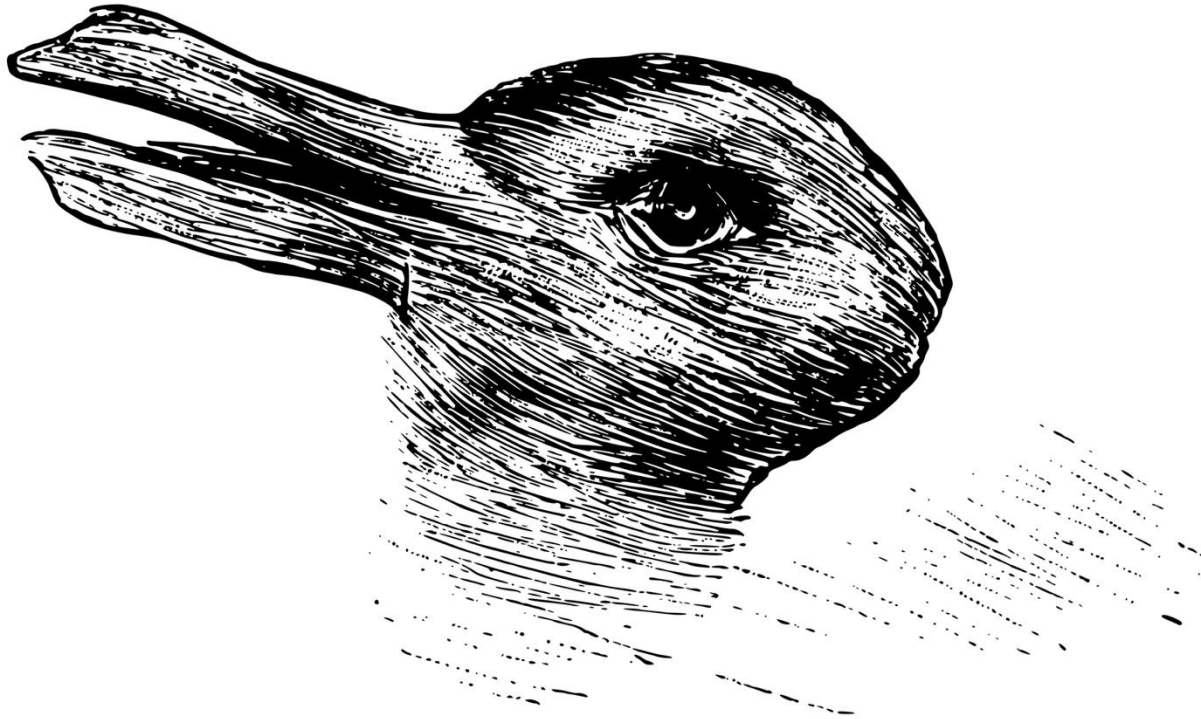


The starting point for philosophical and psychological theories of perception is that the perception of the object is in the observer, and not the object.



Anderson & Winawer, 2007

Welche Tiere gleichen ein- ander am meisten?



Kaninchen und Ente.

For philosophical and psychological theories of perception is that the perception of the object is in the observer, and not the object.

Kant (1781) is reference for the idea that

- sensation arrive many at once, ordered in space and time, but not joined to one another.
- The mind combines these in a synthesis step consisting of operations (a priori's).
- The point of unification is to which the perception belongs (consciousness).

Helmholtz reading of Kant: mind imposes these a priori's on the world. (Marr's reading of vision)

Lorenz reading of Kant: a priori's developed as a response to the structure of the world.

Dominant (Helmholtz) branch

Stimulus input is underdetermined

- Features
- Segmentation
- Integration
- Matching
- Object perception

- Sketch
- 2.5 D
- 3 D vision

Central problem **binding**:

If the object is built from independent features you need a heavy assembler.

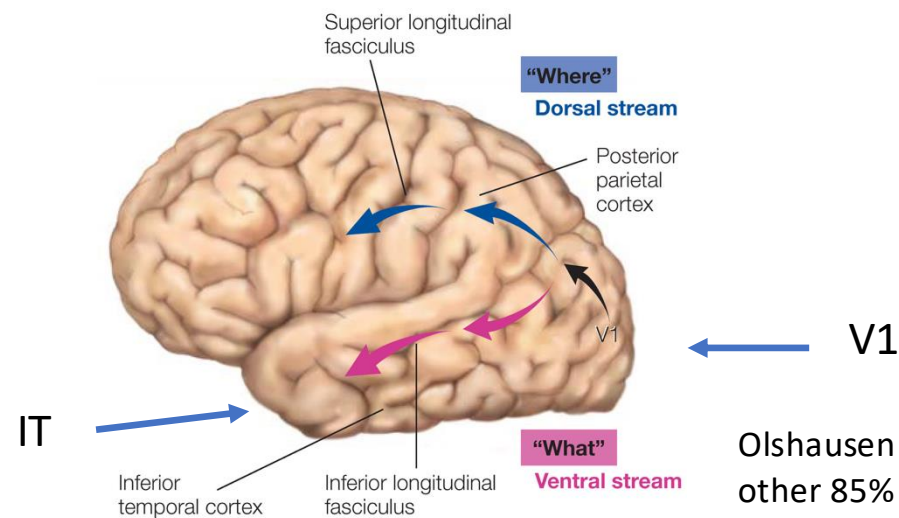
Seeing as reconstructing the world, and act on this reconstruction.

Projecting this on neural activity and organisation

The cells in V1 we record are a biased minority: large, active, responsive to bars. The rest go unreported.

Hand-crafted features explain little of the response to natural images.

- About 15% if the responses in V1.
- There is no substantial number for IT.



Olshausen & Field (2006). What is the other 85% of V1 doing

Dominant (Helmholtz) branch

Stimulus input is underdetermined

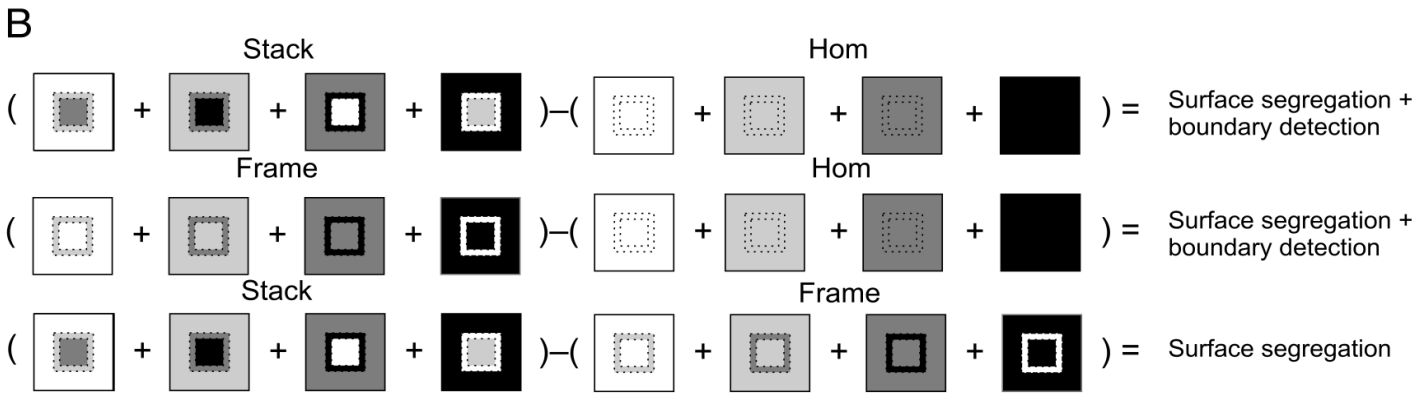
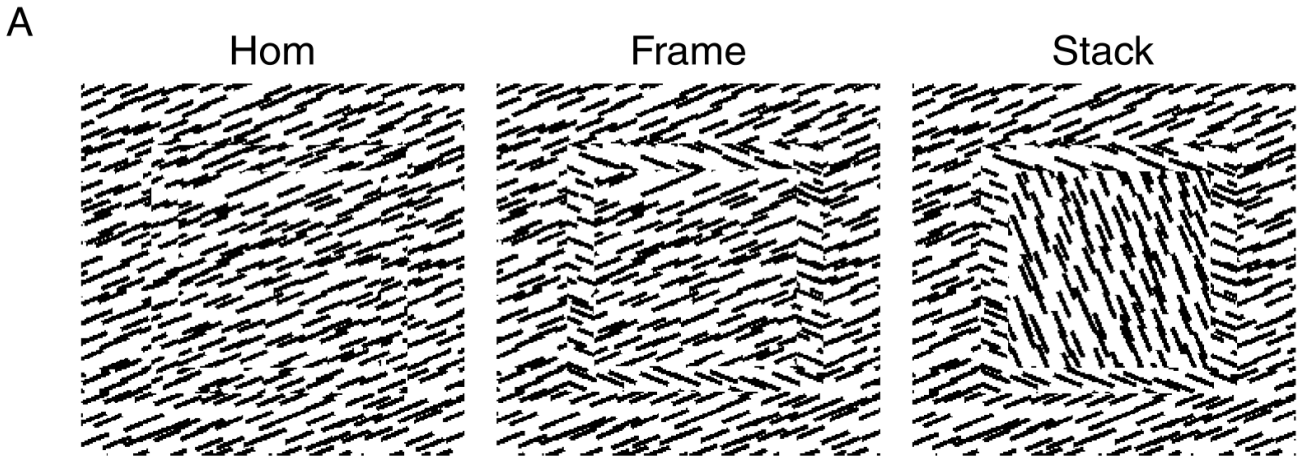
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Central problem **binding**:
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you need a heavy assembler.

Seeing as reconstructing
the world, and act on
this reconstruction.

Very much an example of the
helmholtz branch. Looking for a
function (segmentation) and using
**idealised stimuli & a mind sized
framework**



The resistance (Lorenz) branch

Stimulus input is structured

- Regularities
- Co-occurence
- Relations are in the input
- Affordances

- Direct fit

Central problem: **reach**.
Where do the regularities
stop, and what to do then.

Seeing is acting on the
structure of the world

Dominant (Helmholtz) branch

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Central problem **binding**:

If the object is built from independent features you need a heavy assembler.

Seeing as reconstructing the world, and act on this reconstruction.

At the other hand, the Lorenz branch has produced little in the last century beyond noting the shortcomings of the Helmholtz branch.

This century deep learning has invigorated the Lorenz branch, but is also criticized for being too complex to be useful.

The resistance (Lorenz) branch

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Empirical theory

A mind sized theory presumes a specific world



Human experience as a river

James 1905, 1909

Research is often:

- Dip a bucket in the river.
- You learn what is in the bucket.
- Your theory and your instrument are the bucket.
- Dip again: the same (this is more about reliability).

An ichthyologist casts a net, checks what is in it and declares : all animals in the river are fish larger than 5 cm.

Eddington (1939)

A mind sized theory presumes a specific world



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James 1905, 1909

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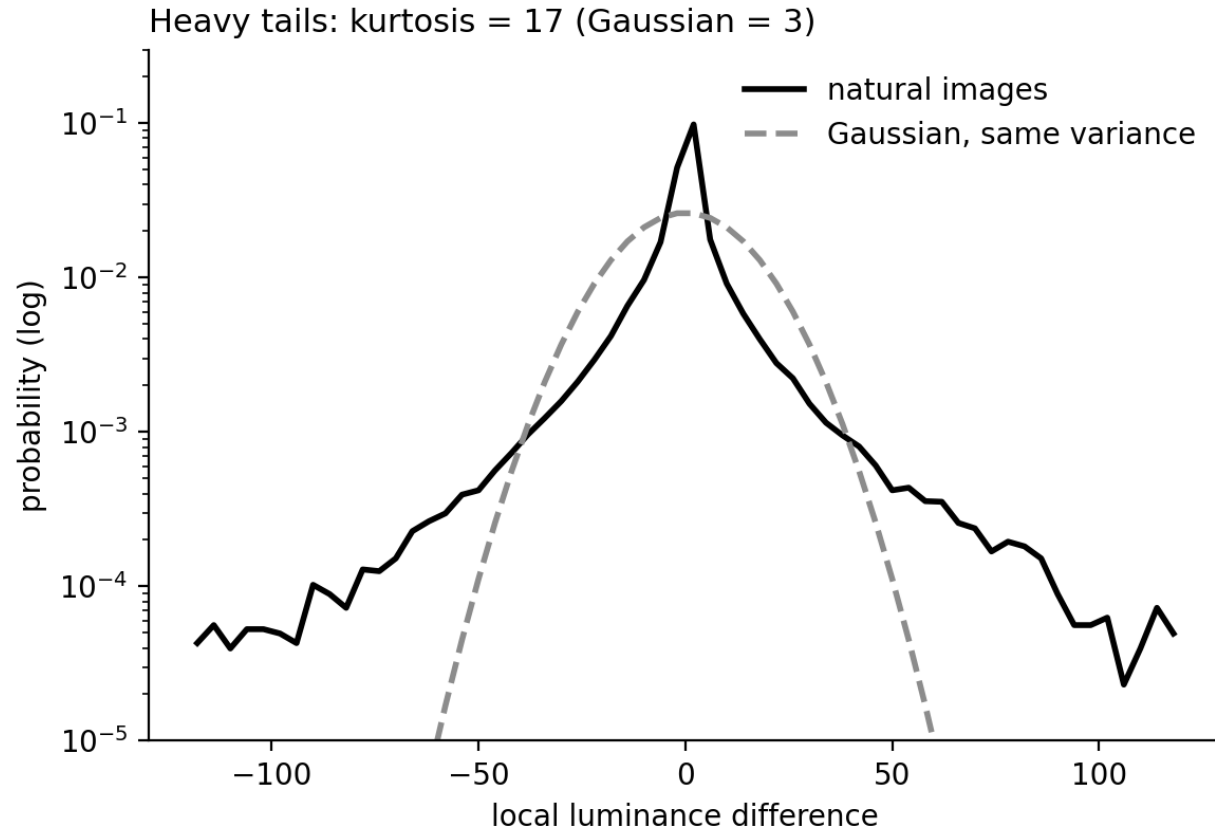
- Dip again: the same (this is more about reliability).

The bucket must be mind-sized, a theory you can hold in your head and you can reason with.

For perception this could be a theory about features, functions and object perception.

It cannot be on 60 million parameters.

The world we live in: structured but not mind sized



Not noise.

Not a rulebook.

At a local level: Enough structure to exploit; too much to hold.

But also:

- Shadows fall away from the window.
- Ceilings are white.

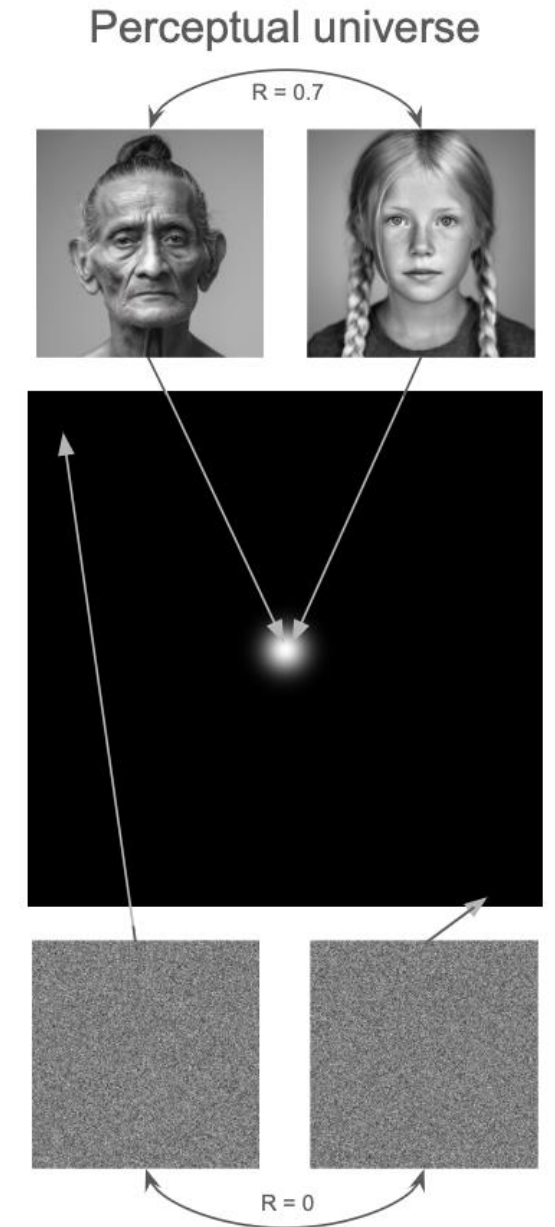
Simoncelli & Olshausen 2001; Geisler 2008.

The world is neither random nor mind-sized



But also:

- Shadows fall away from the window.
- Ceilings are white.
- Table's are on the floor.



The mind is local

We are great in social situations.

We describe atoms in the same terms as the solar system because we can handle that.

Quarks



Atoms

Molecules

Cells



Organisms

Groups



Societies

Ecosystems



Planetary



Increasing distance of understanding

The world is neither random nor mind-sized

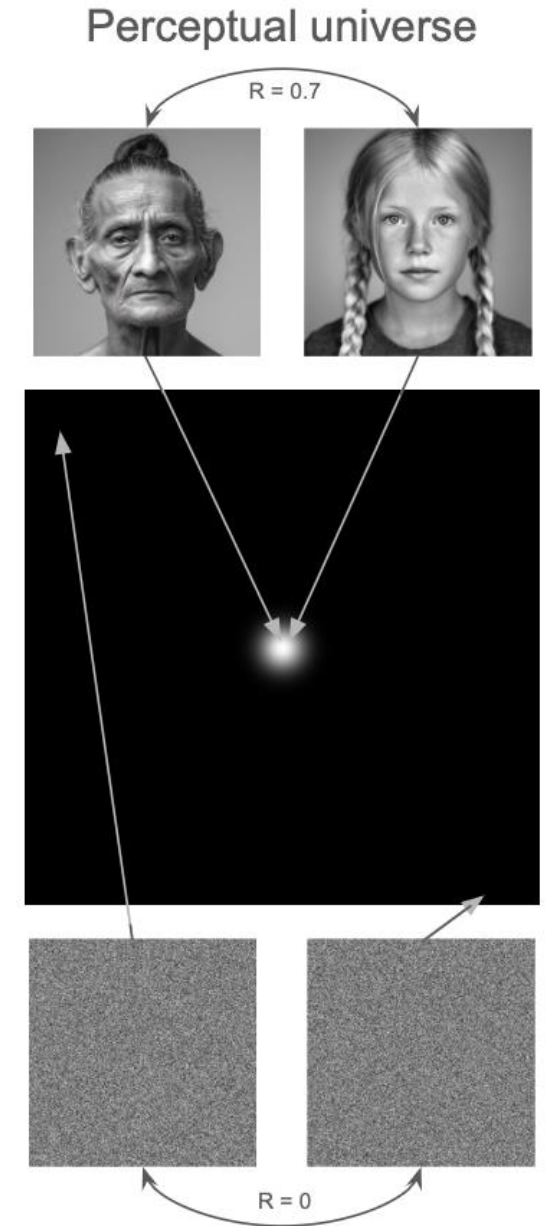
The world we live in is in between: structured, correlated, heavy-tailed, and not summarised by anything much smaller than itself.

We evolved to survive in it, not to understand it. The mind is a tool for getting through a day, not for writing the theory of one.

So the compression fails from both ends. The world we live in does not fit a small model, and the mind was not evolved to hold a large one.

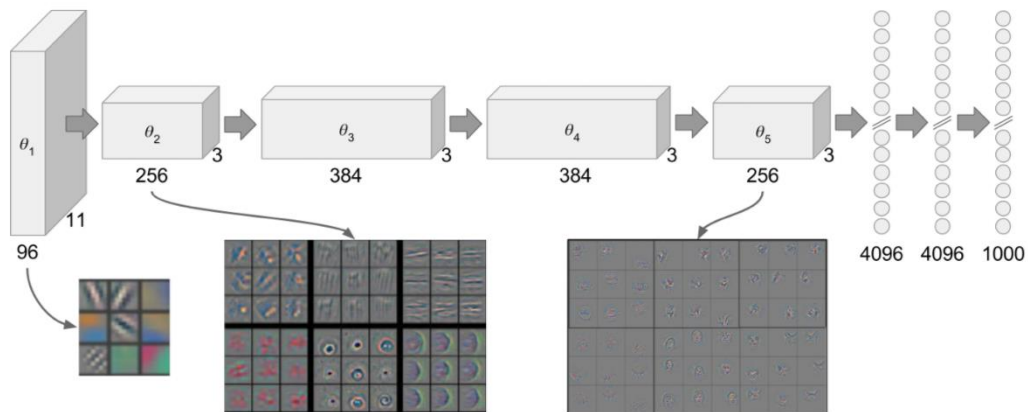
Scholte, H.S. & de Haan, E.H.F. (2025). Beyond binding: from modular to natural vision. *Trends in Cognitive Sciences*, 29(6), 505–515.

Scholte, H.S. & de Haan, E.H.F. (2026). Beyond binding: lunch can be surprisingly affordable. *Trends in Cognitive Sciences*, 30(4), 290–291.



Mind-sized vs world-sized

	Parameters
Mind sized (taken to be high at understanding)	Narrative theory0
	Reinforcement learning models2 to 10
	Drift diffusion4 to 12
	Hierarchical Bayesian100 to 1,000
World sized (taken to be high at predicting)	Deep neural networks25 million and up



deep convolutional neural network models

Is this a mechanistic
model of the brain?

Animals, and animal models of a task or disease are high multi-parameter models that are, on account of our shared origin, considered to be interesting for understanding human behavior



Fantastic dnmals and where to find them

Scholte (2018)

- Artificial agents can be trained in an environment that is similar to biological agents.
- Artificial agents can be / have an architecture that is similar to biological agents.
- Artificial agents can be trained to get behavior that is similar to biological agents.
- Therefore artificial agents could be used in a same way as animal models in research.
- Lets call them **Dnmals (a DNN animal)**.

Fantastic dnmals and where to find them

Scholte (2018)

Advantages of Dnmals over Animals

- All elements are known and can be probed (contrary to for instance a mouse).
- Can be raised with a specific diet of information.
- They work for free (but cost energy).
- Few if any motivational problems.
- No (known) ethical problems with using breeding, culling or lesioning of DNNs.

Using DNNs as animal models and not principled mechanistic models allows a type of research that is more flexible.

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An approach

1. Models

Find, build or train several models that show the behaviour.

Several. One model describes itself.

2. Experiment

On them: lesion, vary depth, retrain, track training.

With them: as instruments.

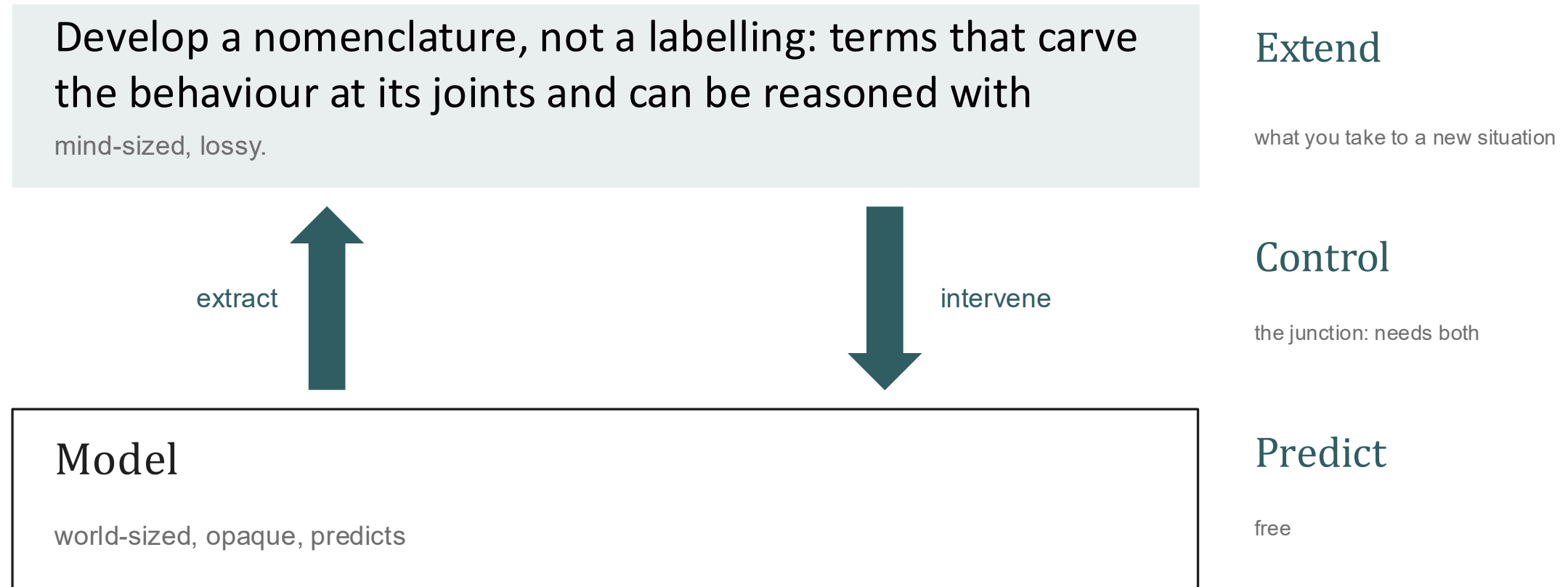
Extract a nomenclature. A name for a variable.

3. Check

In the model, by interventions the name predicts.

Adequate, not correct. Adequacy has a degree.

Not a pipeline. Two levels, kept together.



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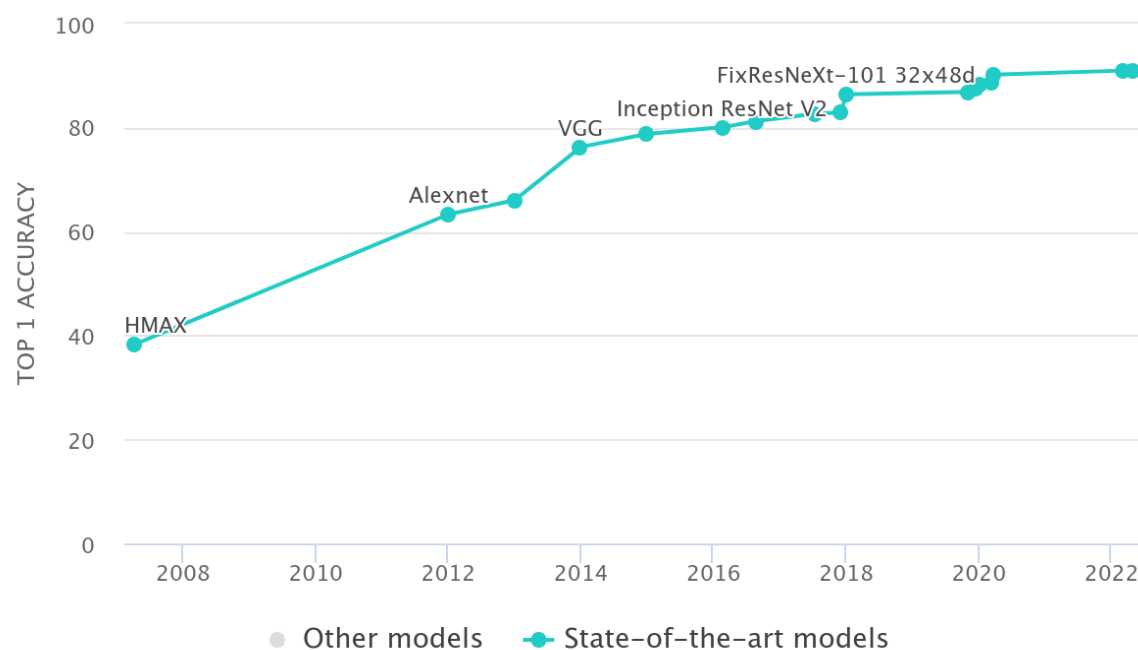
A shared world, not a shared objective

Empirical theory

What we believe about DCNNs and the brain

1

DCNNs do object recognition well.

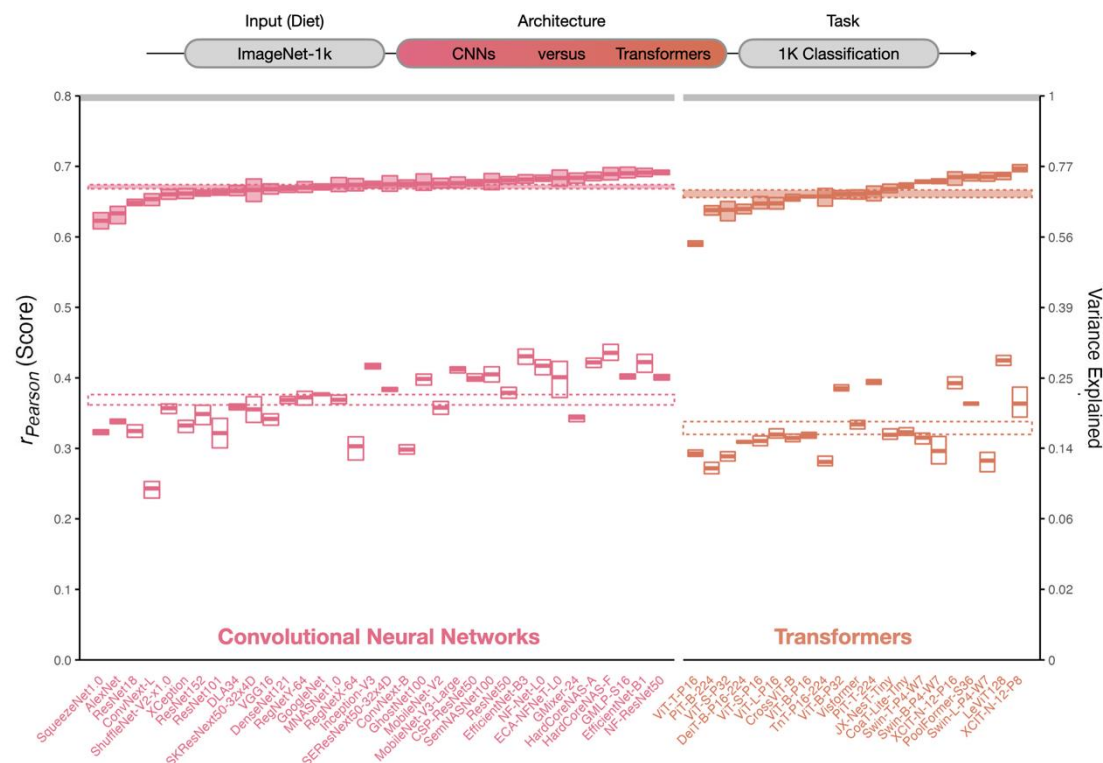


Or at least capable of labelling an image

What we believe about DCNNs and the brain

- 

- 



Conwell et al., 2023

What we believe about DCNNs and the brain

1

DCNNs do object recognition well.



2

DCNNs predict brain activity well.



3

They predict brain activity because the process of object recognition in the brain aligns with the process of object recognition in a DCNN (Yamins et al., 2014; Schrimpf et al., 2020).



Bowers et al (2023) have made the point this does not logically follow.

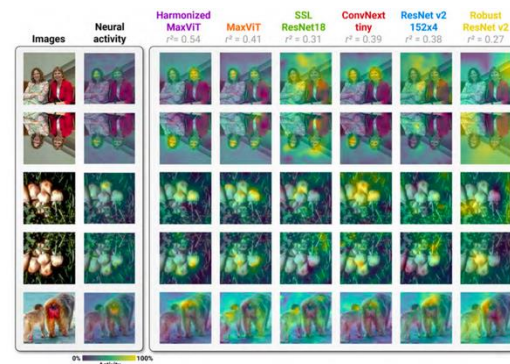
However, improvements on DCNNs models show worse alignment with neural activity at a single unit level.



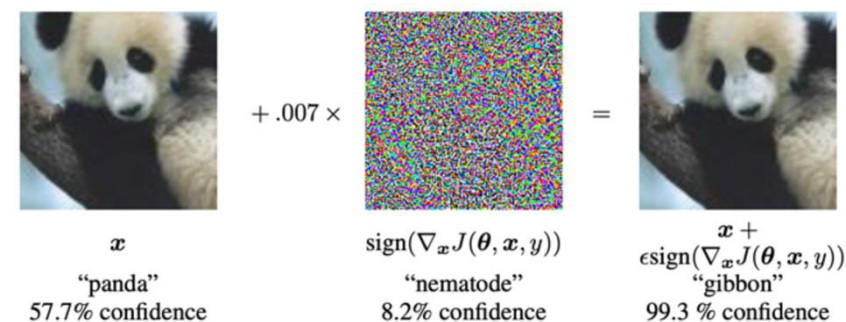
Deep Convolutional Neural Networks

However, improvements on DCNNs models show worse alignment with neural activity at a single unit level.

DCNNs are easily fooled by small perturbations (Goodfellow et al., 2015).



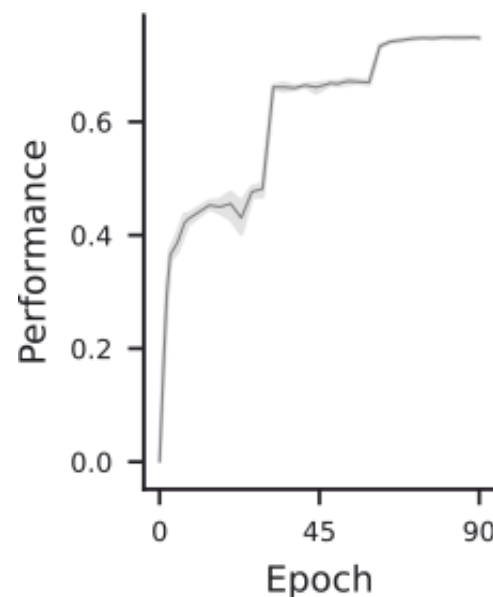
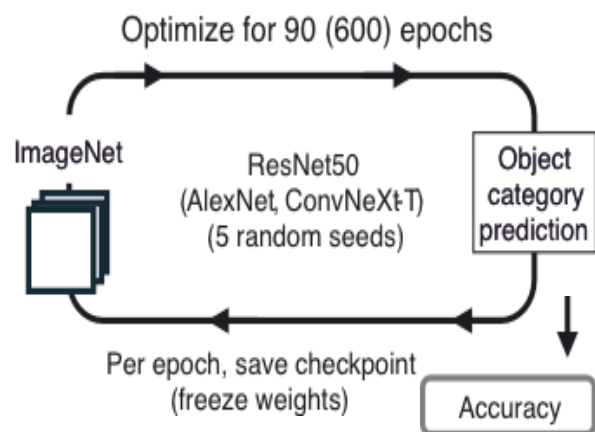
Linsley et al., 2023



Goodfellow et al., 2015

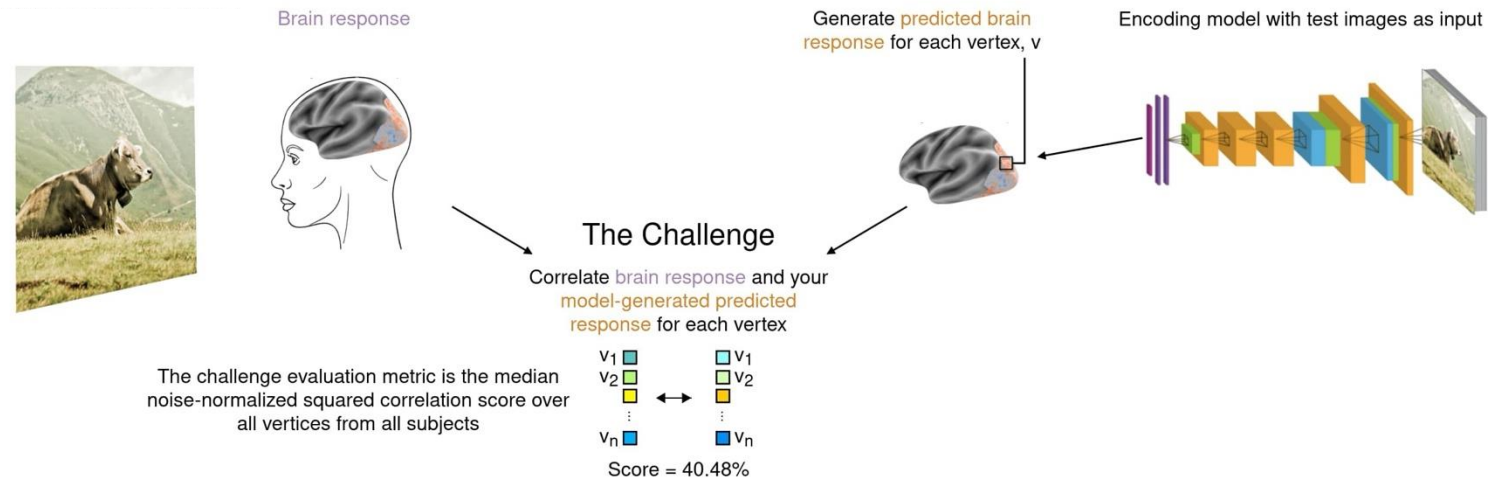
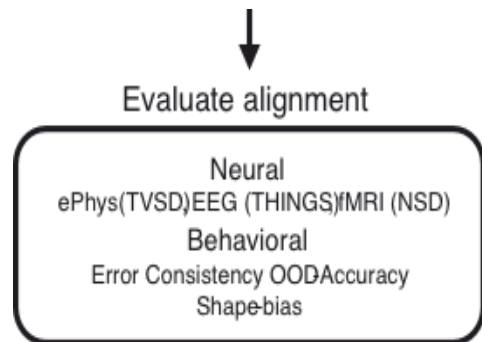
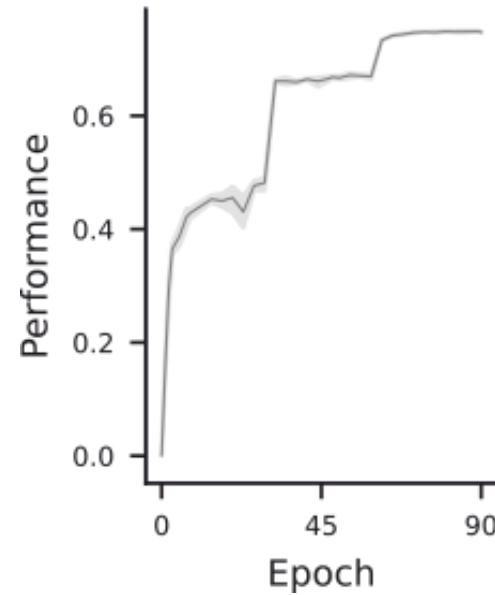
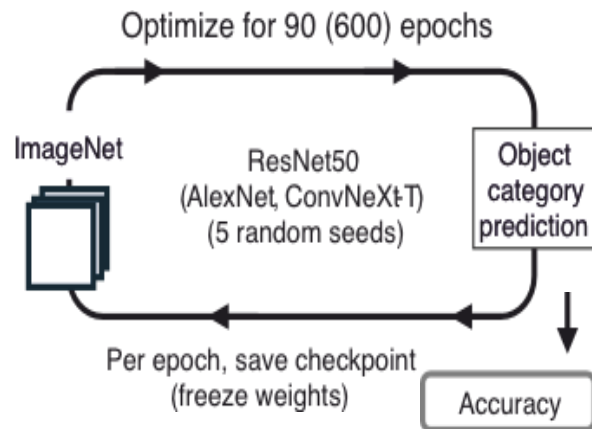
In this talk will explore what brain alignment measures

The relation between the developing DeepNet and brain alignment.

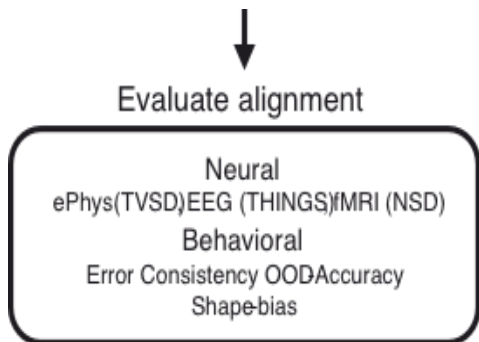
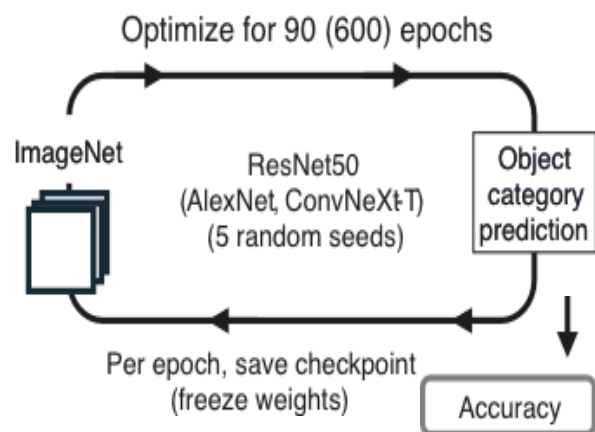


- Deep nets are (typically) trained by repeating the information to be learned many times (an epoch).
- If brain alignment is related to object recognition it should develop with the training of the deep-net.

The relation between the developing DeepNet and brain alignment.



How well does your model predict the human brain's response?

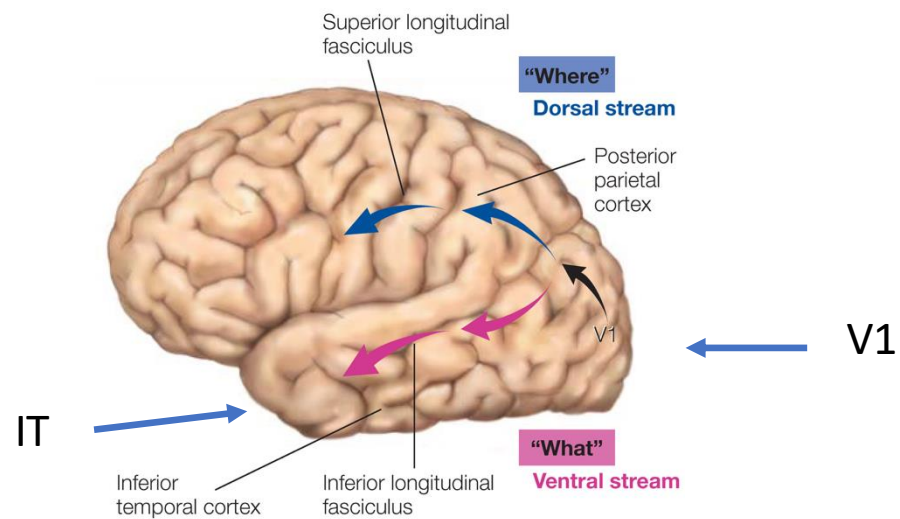
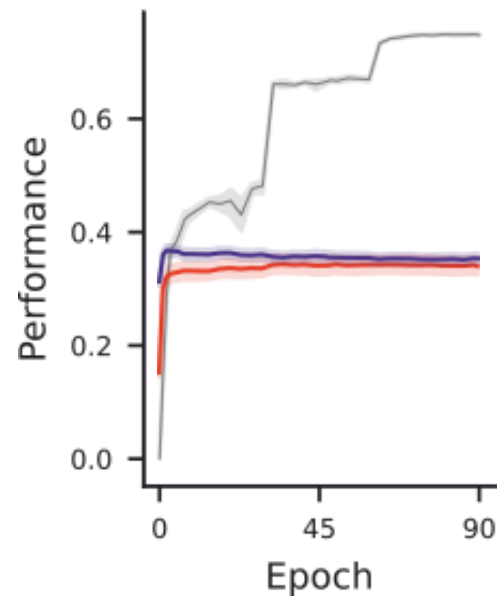


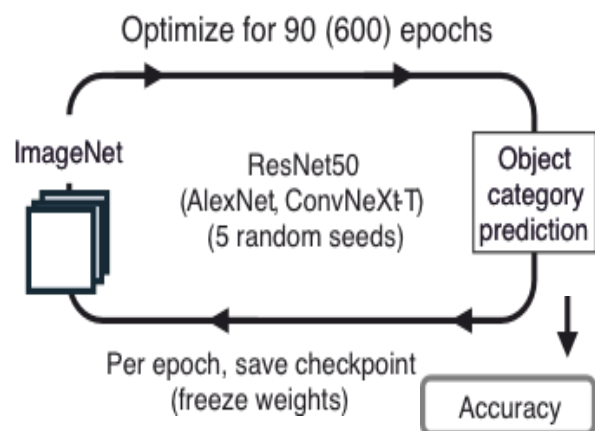
ePhys (TVSD)

— Accuracy (%)

— V1 (r)

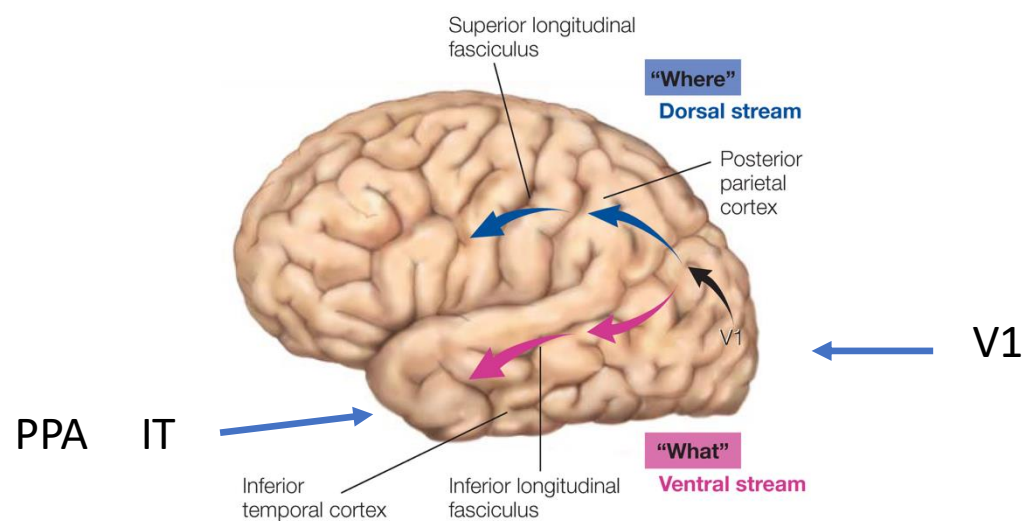
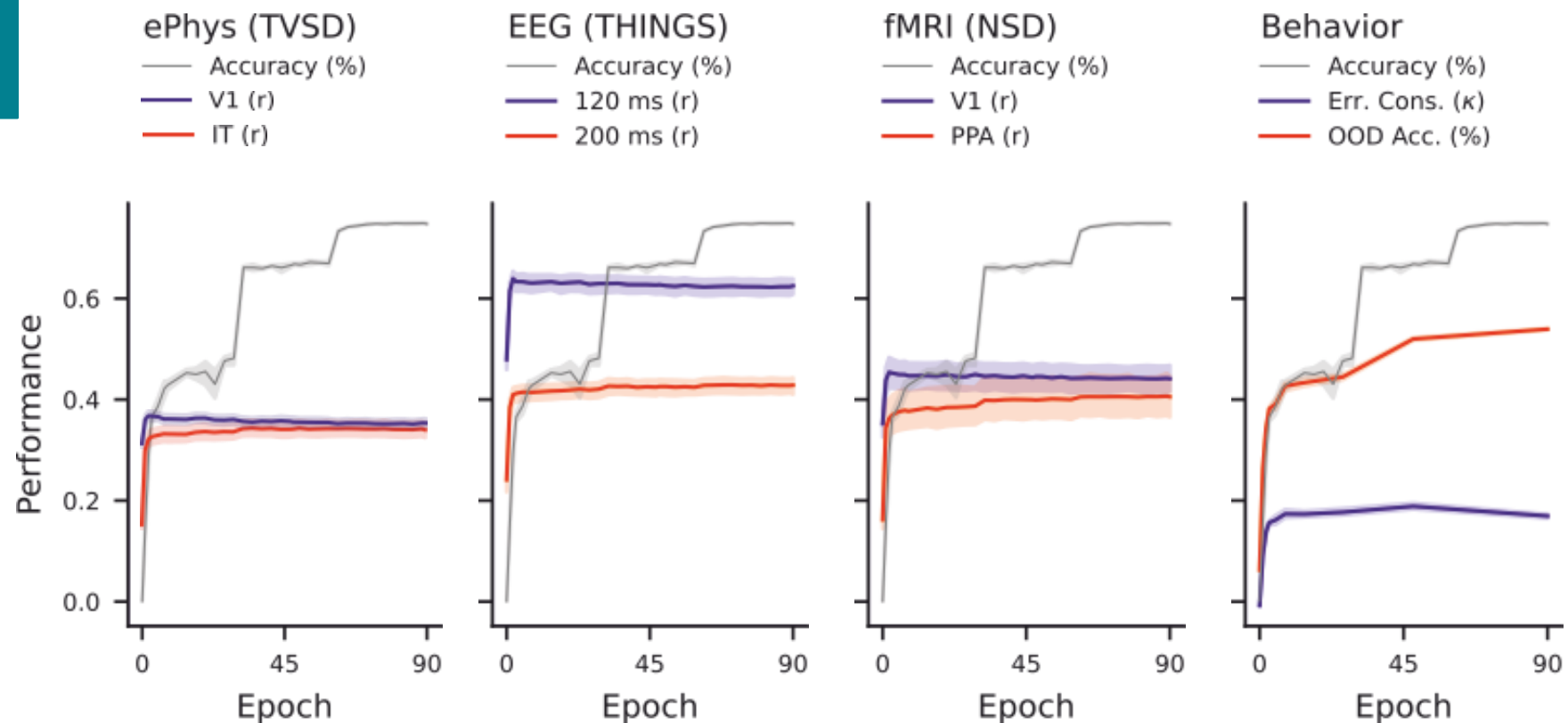
— IT (r)





Evaluate alignment

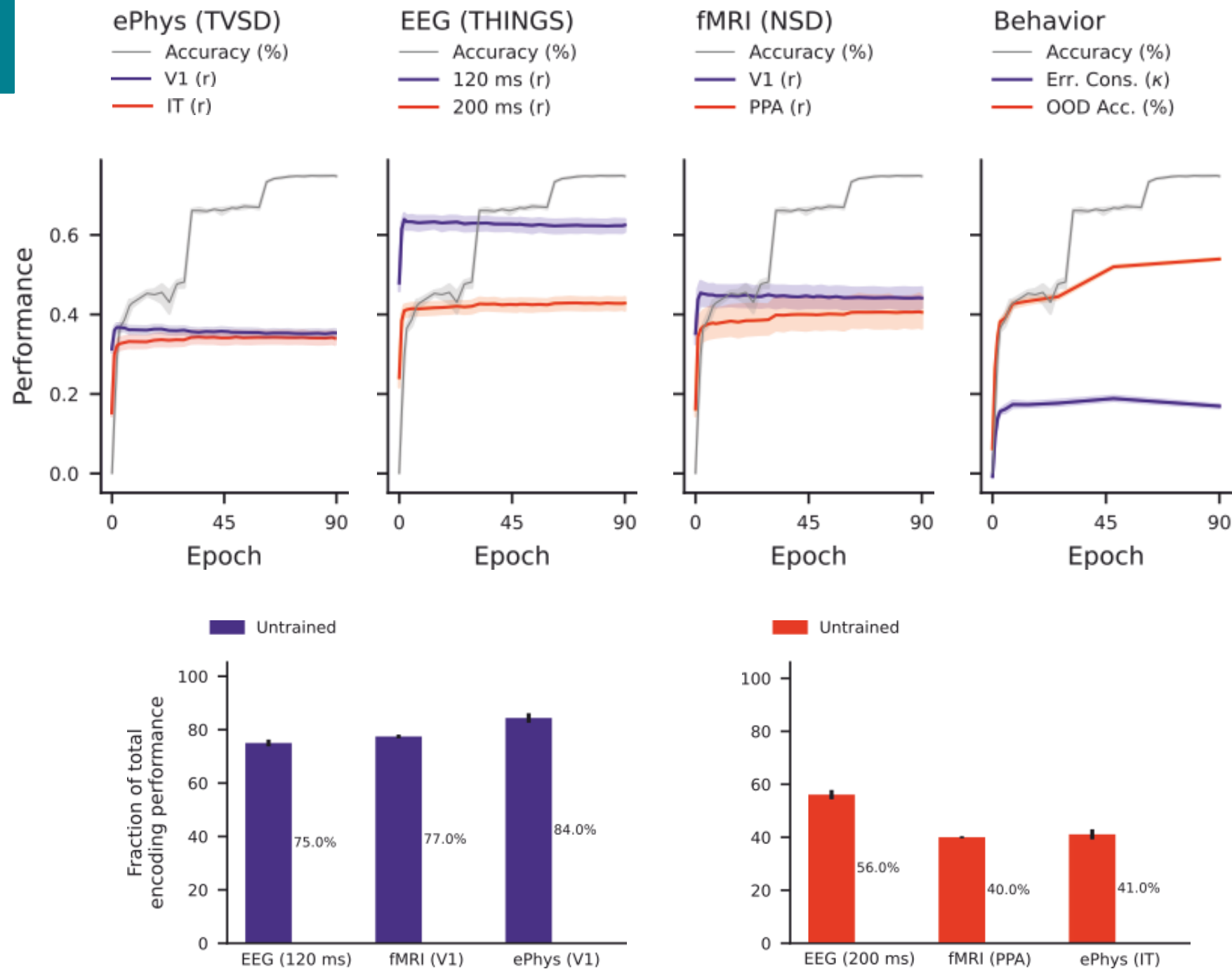
Neural
ePhys(TVSD) EEG (THINGS) fMRI (NSD)
Behavioral
Error Consistency OOD Accuracy
Shape-bias



Seems sensible to divide
DCNN – brain alignment in 3
stages

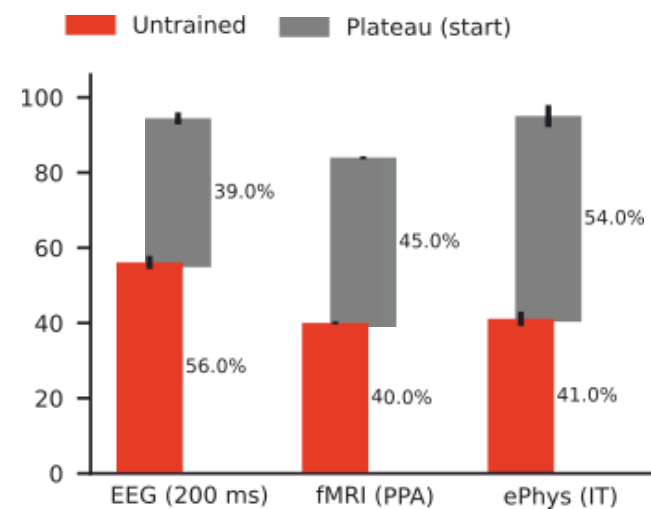
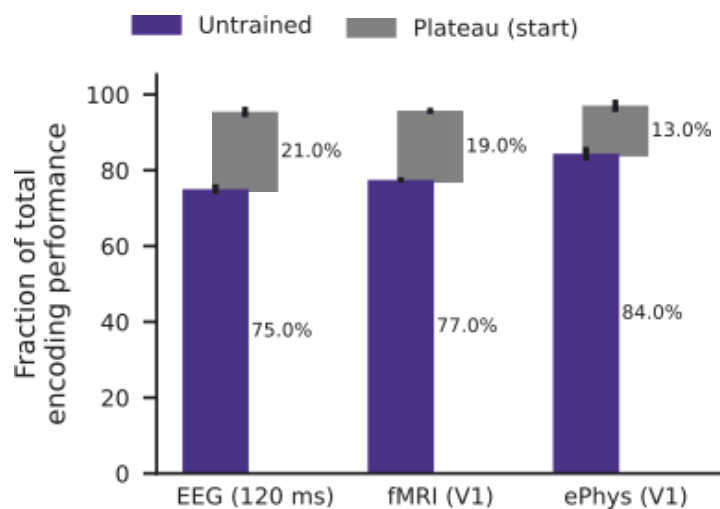
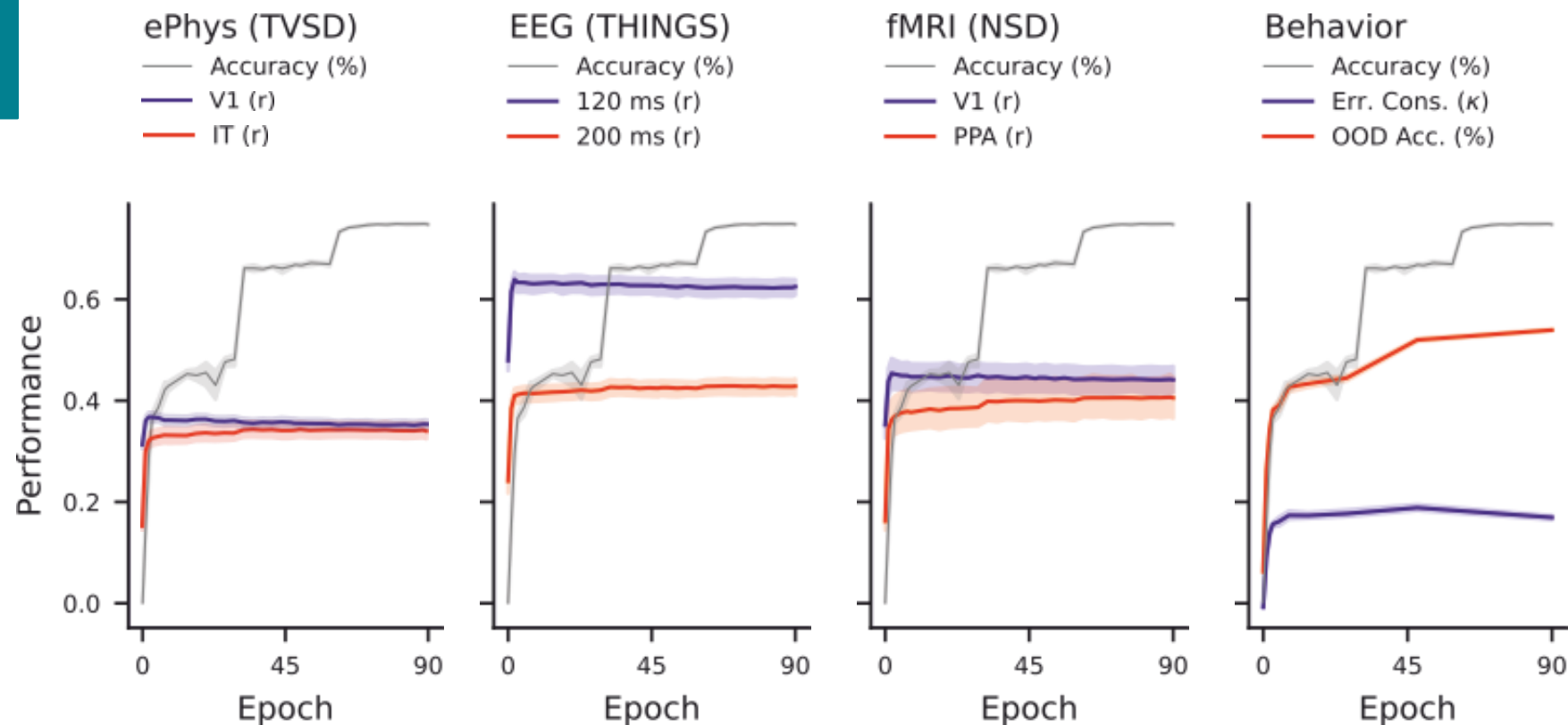
- Initial alignment:**

Cichy et al 2016. Bonner et al. 2026



Seems sensible to divide
DCNN – brain alignment in 3
stages

- **Initial alignment:**
Cichy et al 2016. Bonner et al. 2026
- **Rapid increase**
~first 3 epochs

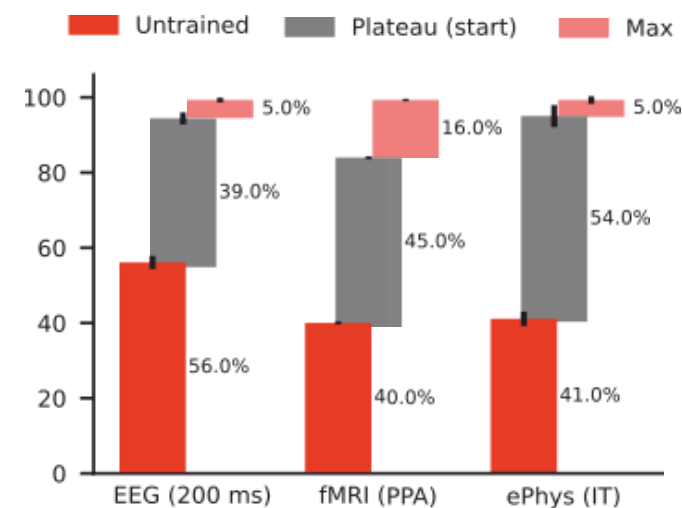
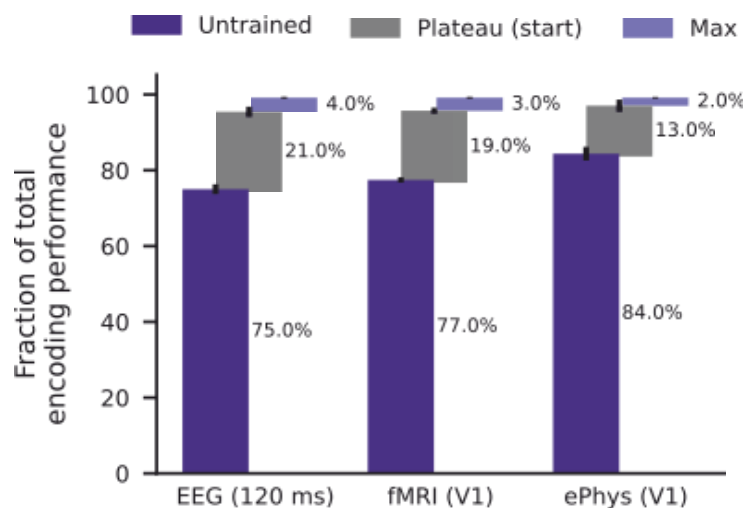
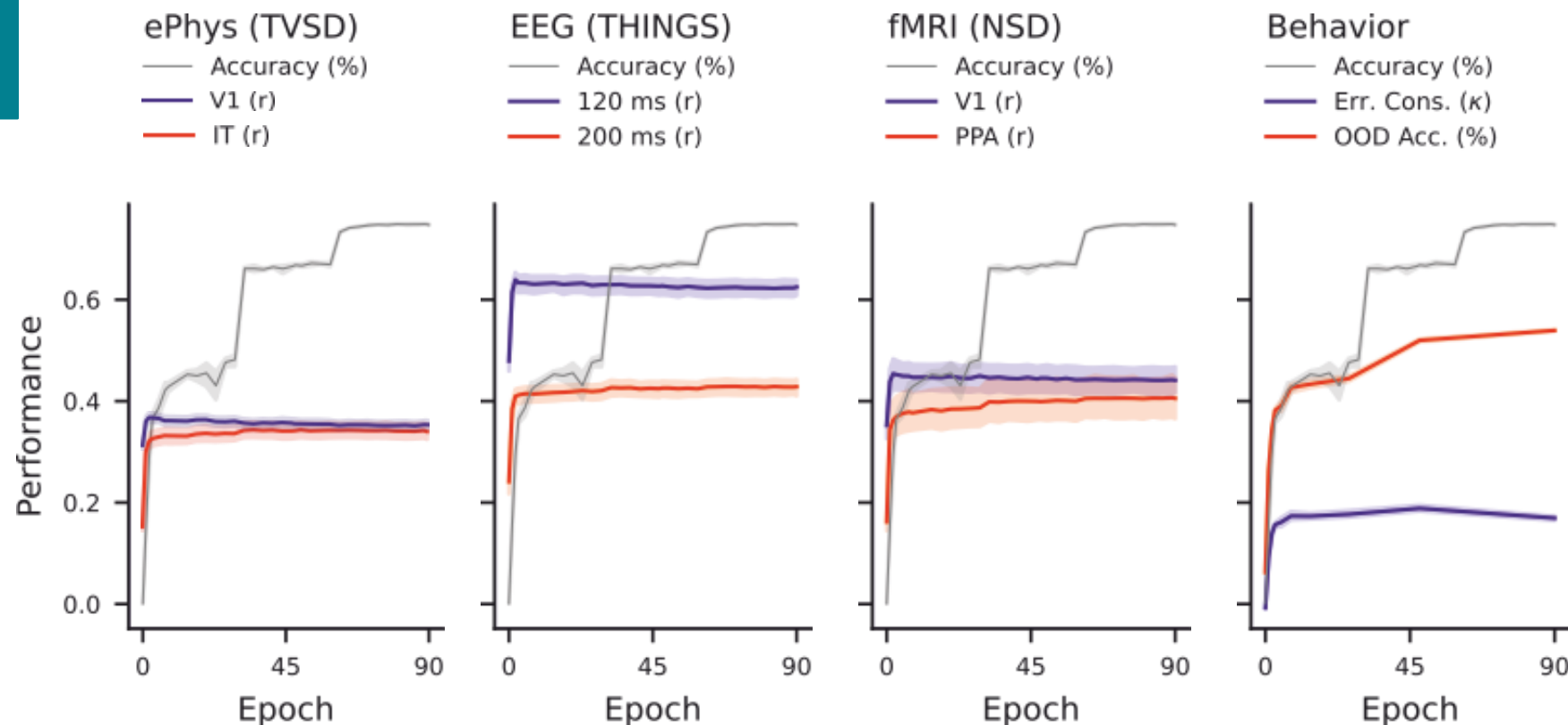


Seems sensible to divide
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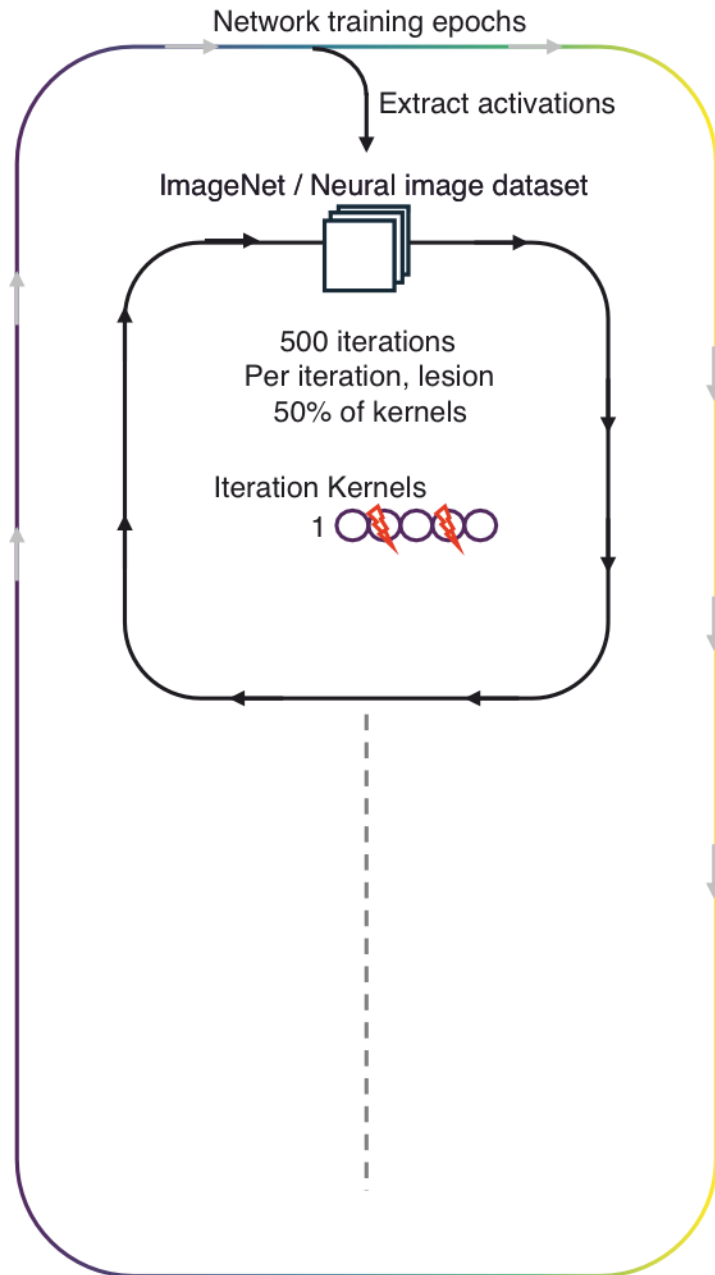
- **Initial alignment:**
Cichy et al 2016. Bonner et al. 2026
- **Rapid increase**
~first 3 epochs
- **Plateau**
Changes within the SEM

Conclusion

Alignment is nearly complete when
classification has barely begun: the
task-optimization prediction largely
fails (there is some increase,
especially in PPA).



Kernel lesioning

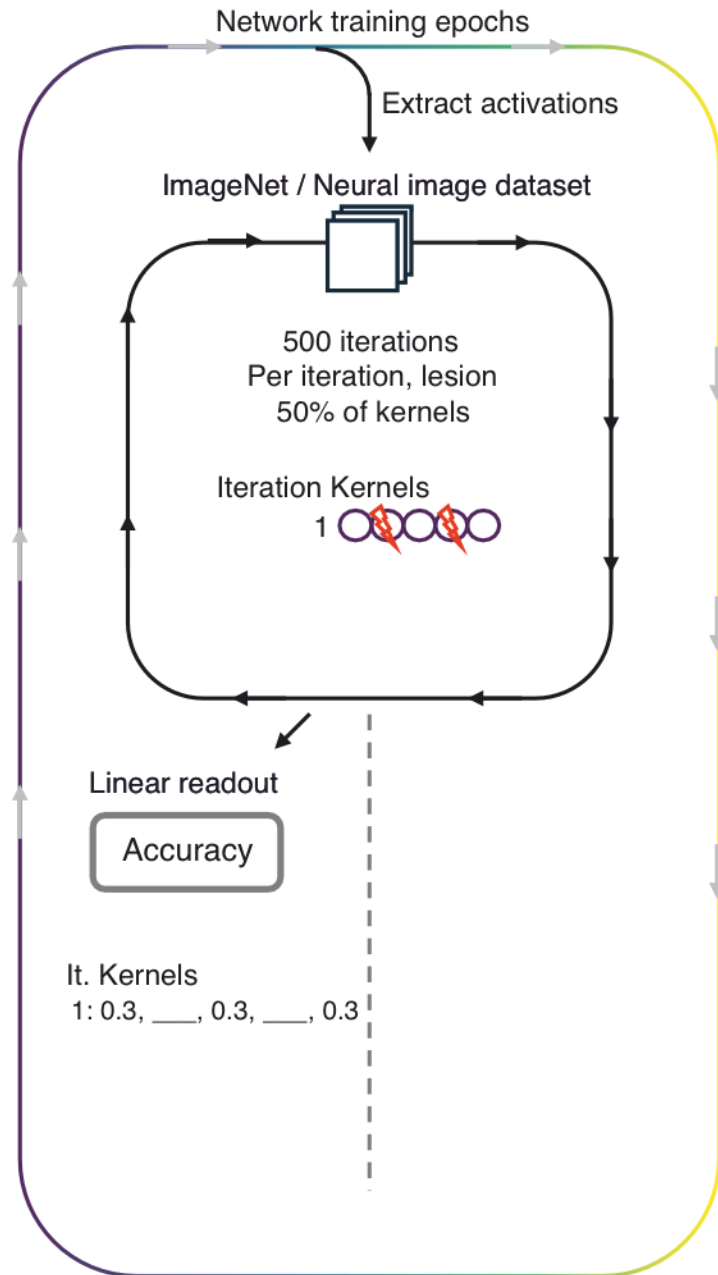


The dynamics suggest brain alignment and object recognition are two processes.

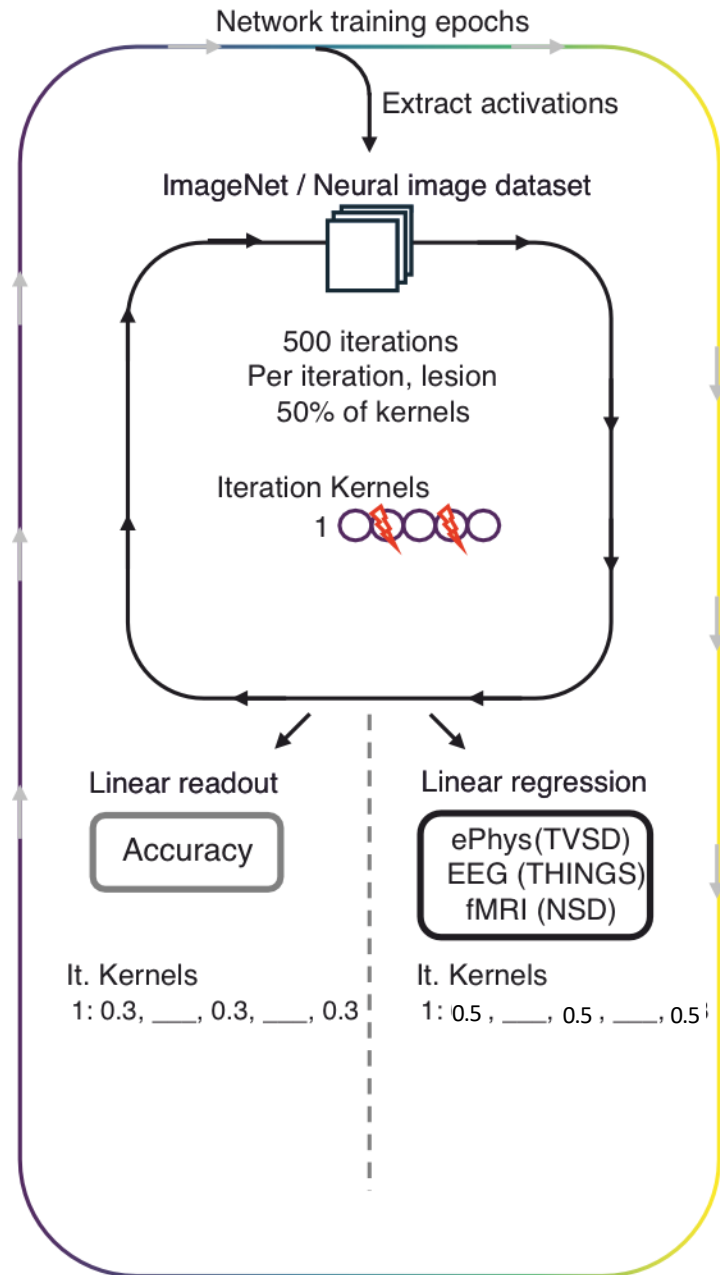
Do they use the same features of the network?

For this we have applied a, we believe, in this form novel way, of analysing the data.

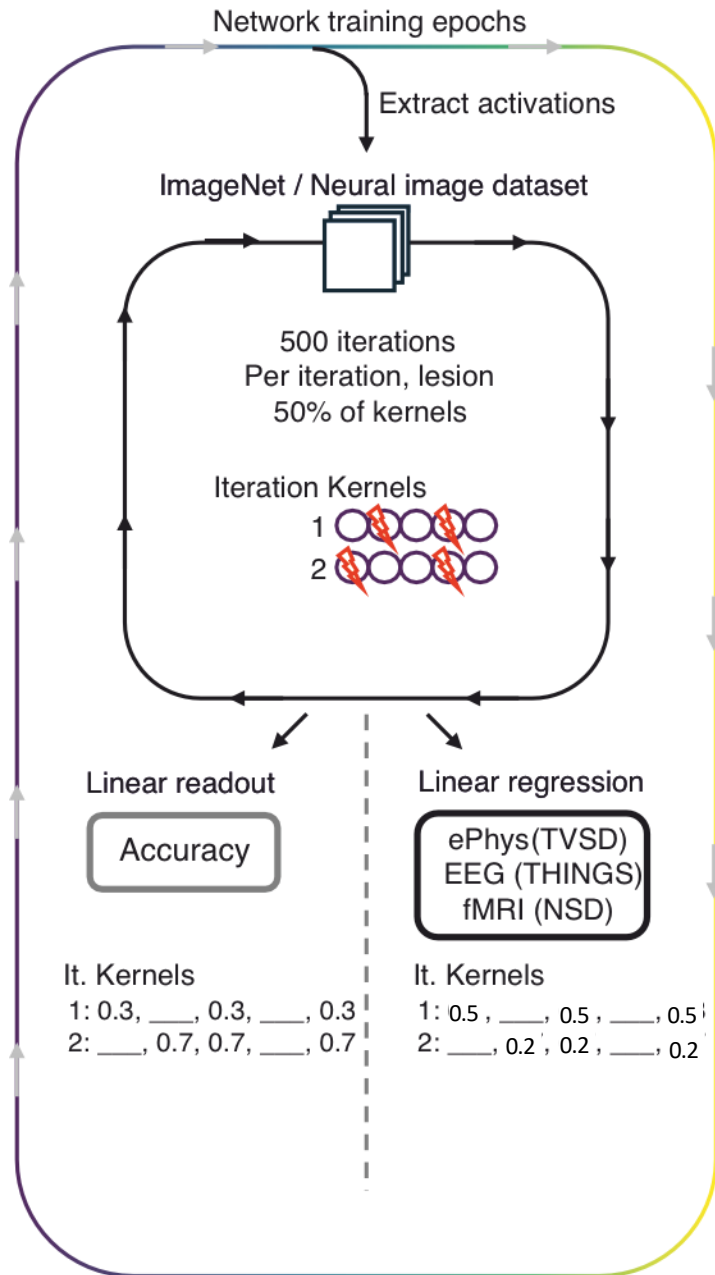
Kernel lesioning



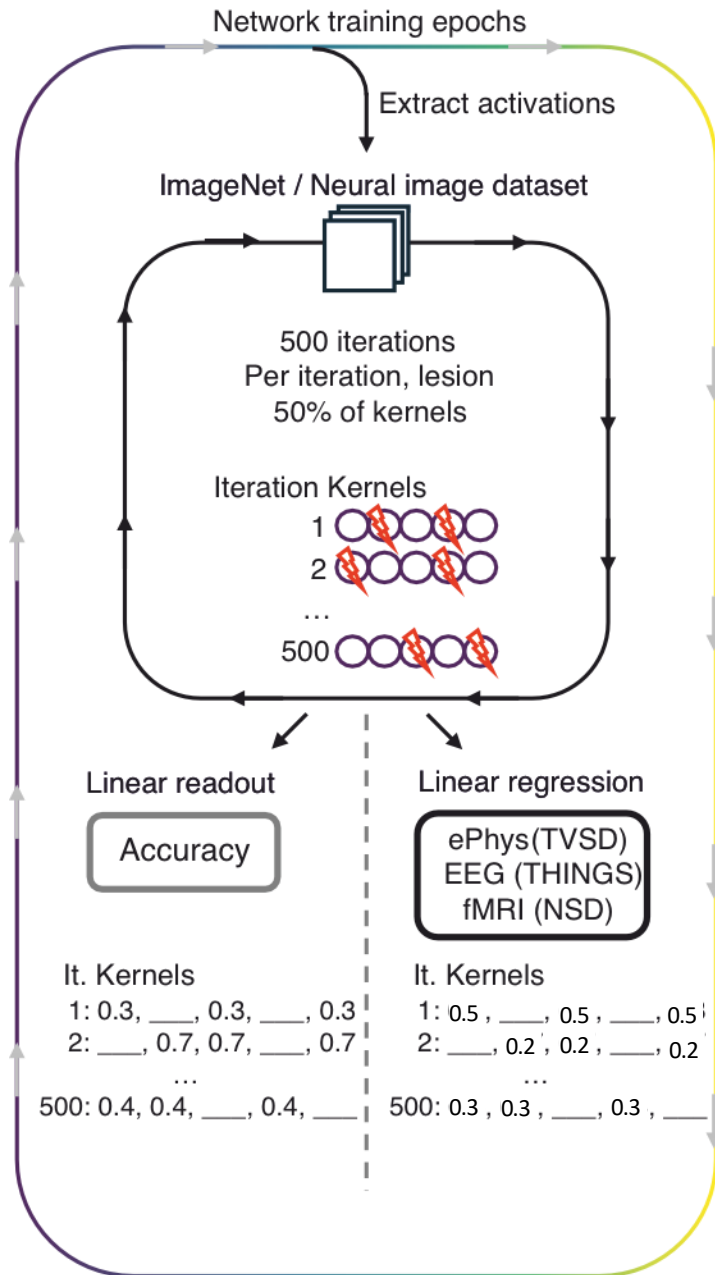
Kernel lesioning



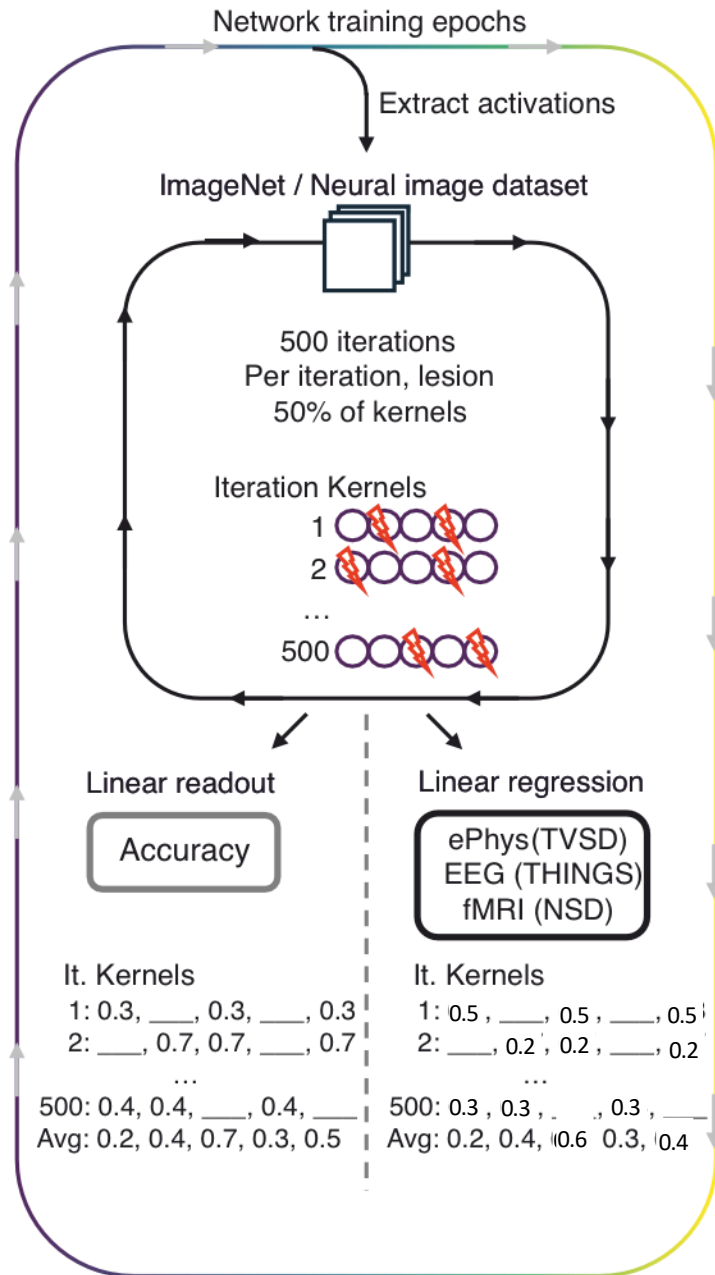
Kernel lesioning



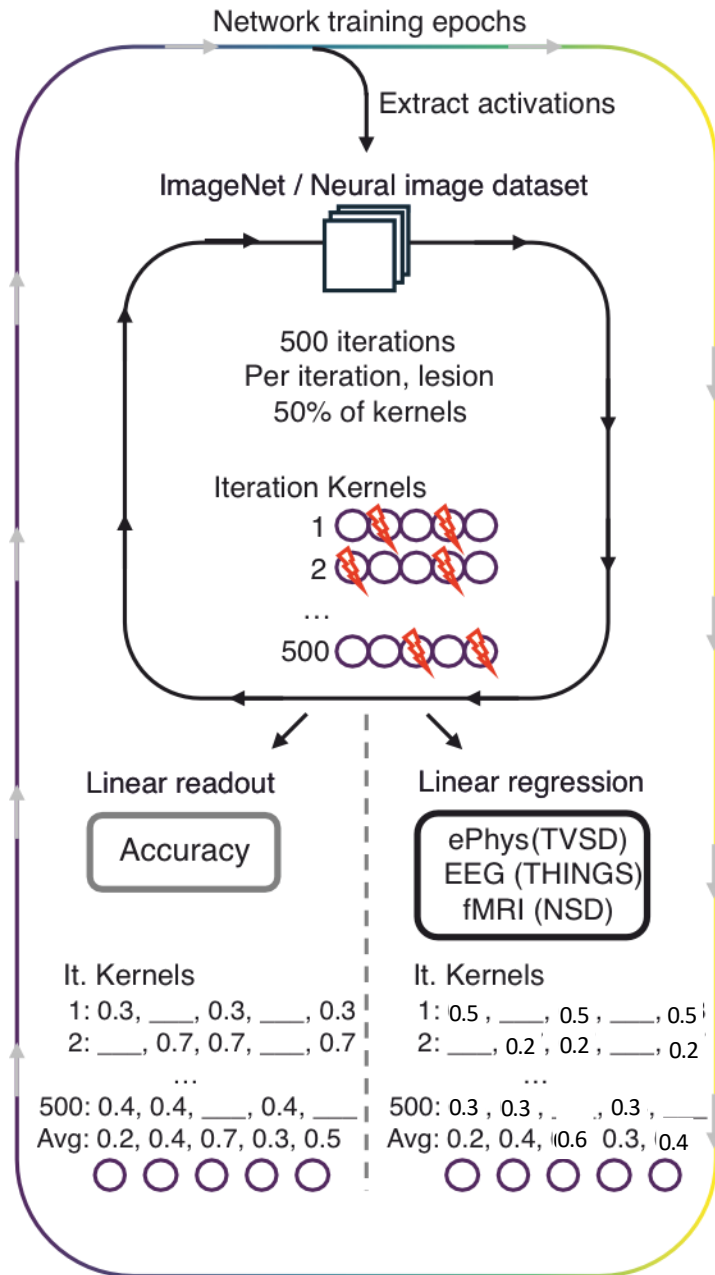
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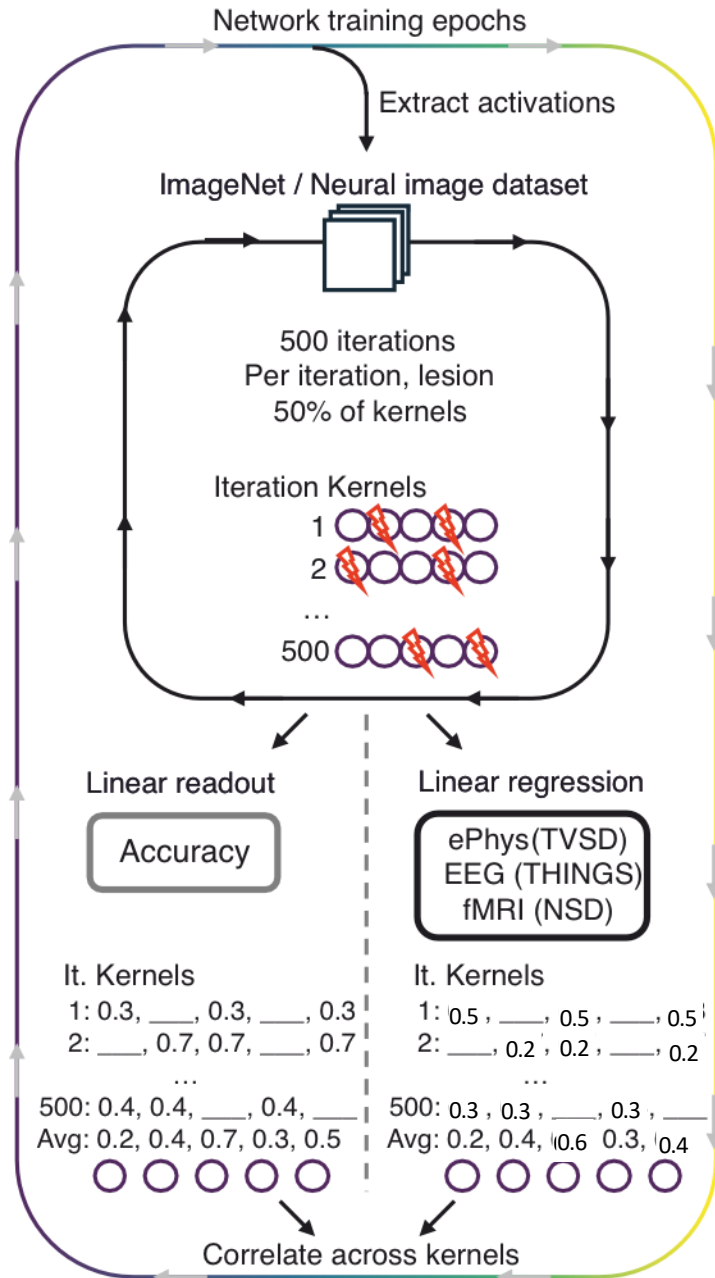
Kernel lesioning



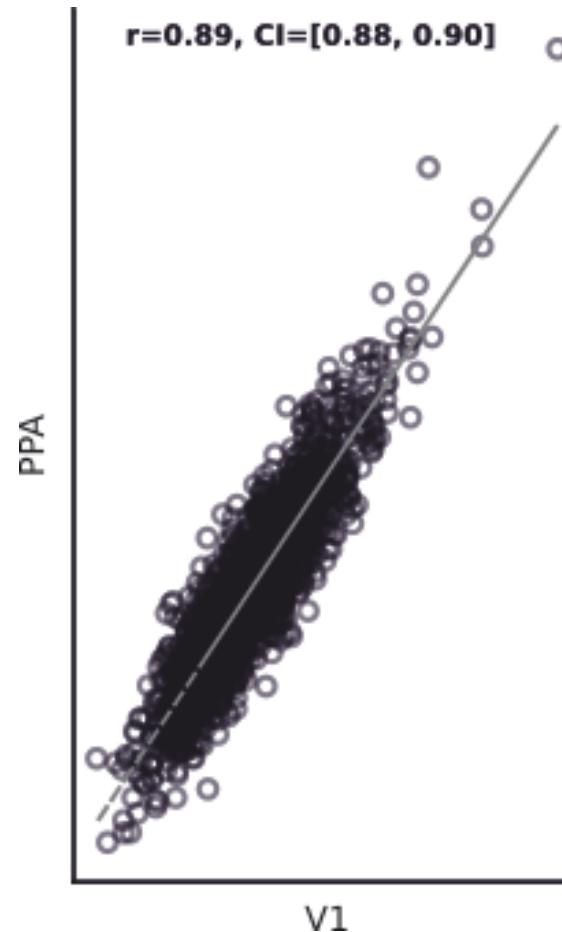
Kernel lesioning



Kernel lesioning



First, a positive control:
do the kernels that help predict PPA
also help predict V1?

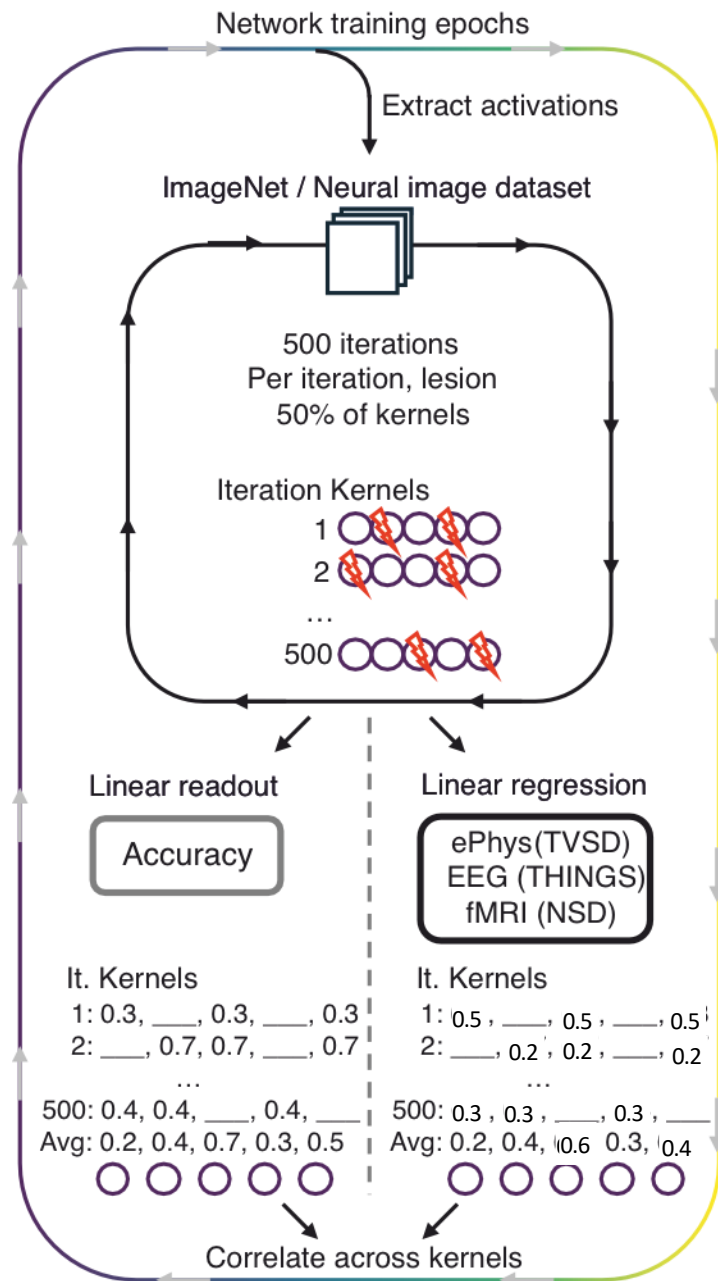


Yes: kernels that predict one region
predict the other. The method finds
shared features when they exist.

So we have a method that measures
each kernel's contribution to a process,
and detects when two processes share
kernels.

Per-kernel contributions estimated as
Banzhaf values (Banzhaf, 1965) via
randomized lesioning (Petsiuk et al.,
2018); task-contribution framework:
Scholte et al. (2018)

Kernel lesioning



ePhys (V1)

EEG (120 ms)

fMRI (V1)

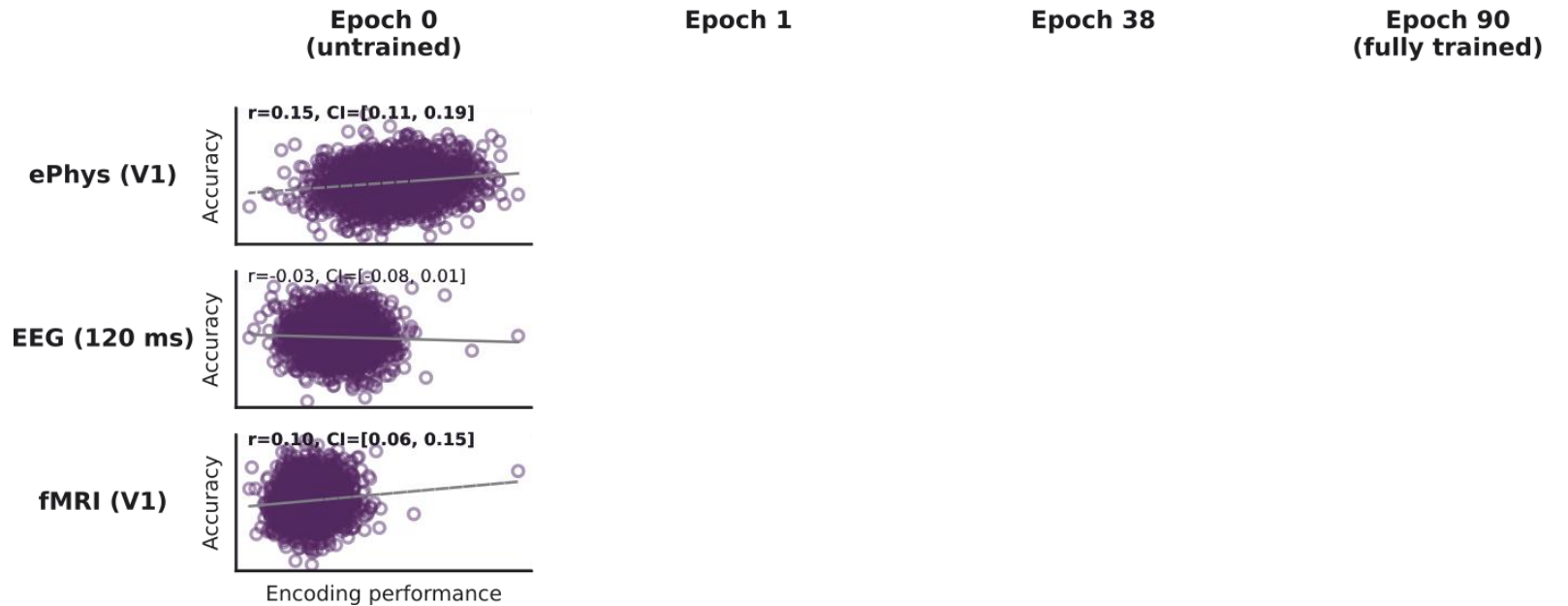
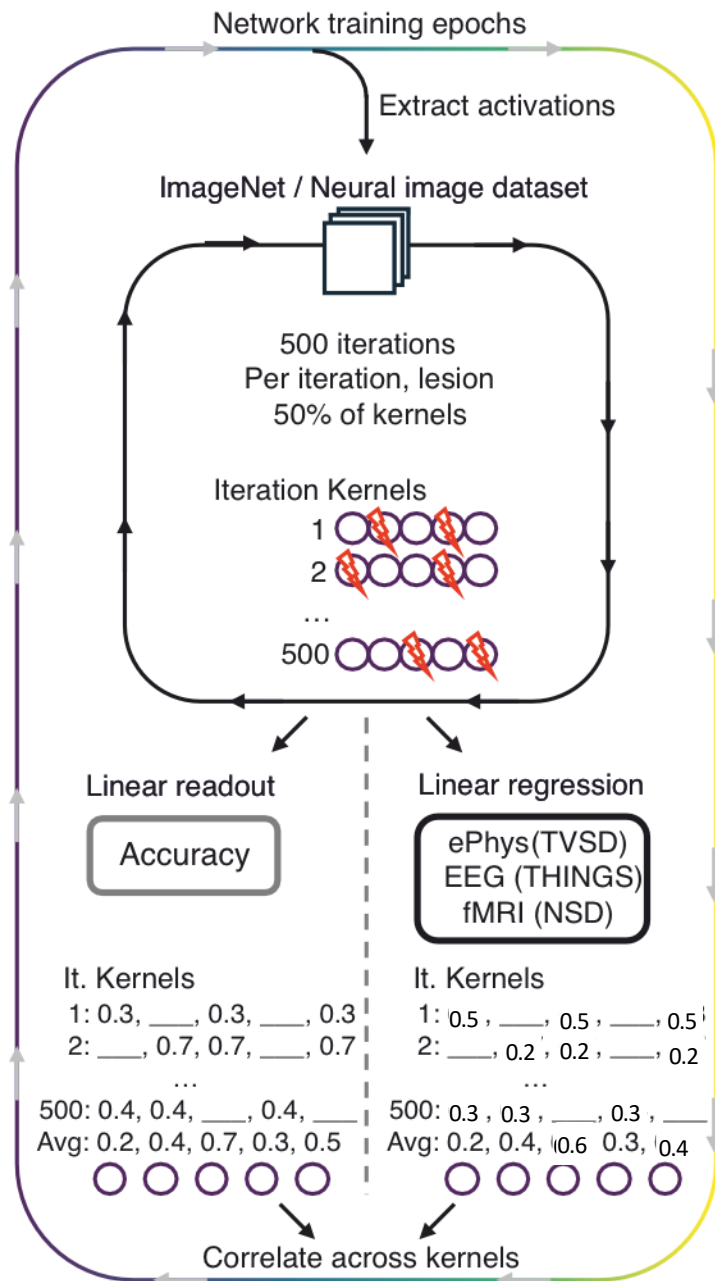
Epoch 0
(untrained)

Epoch 1

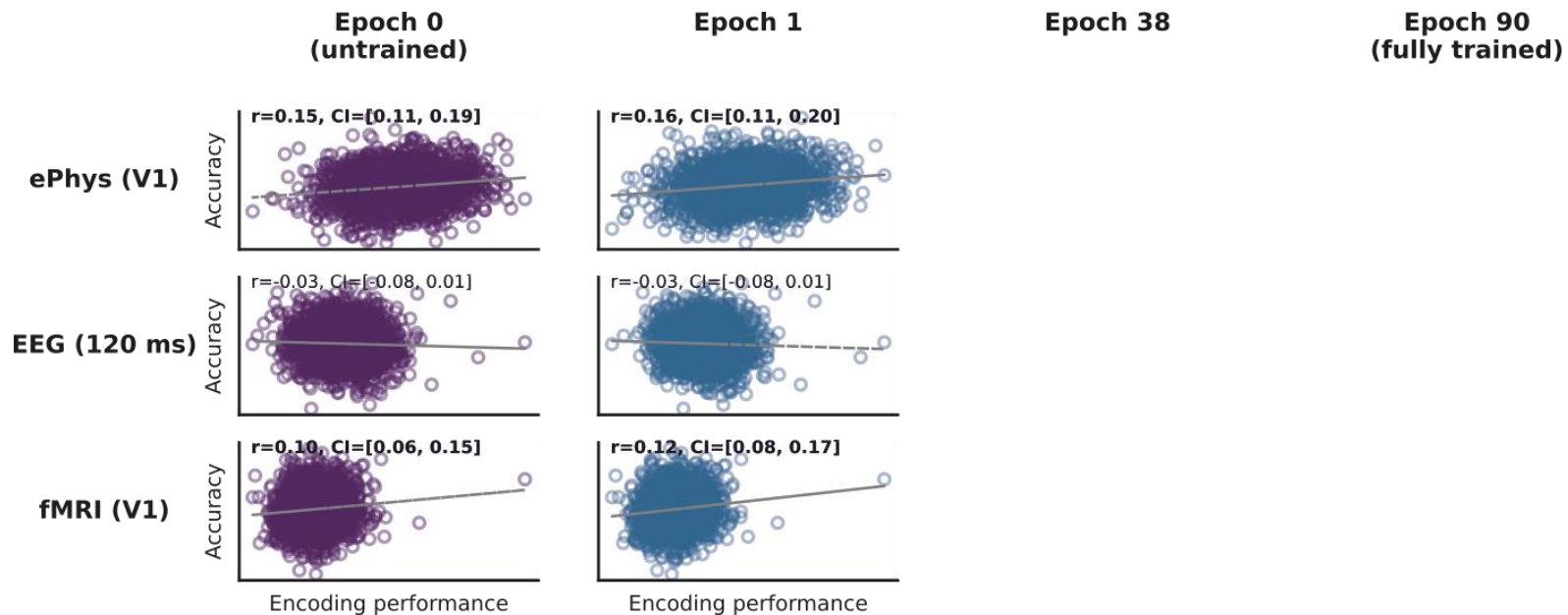
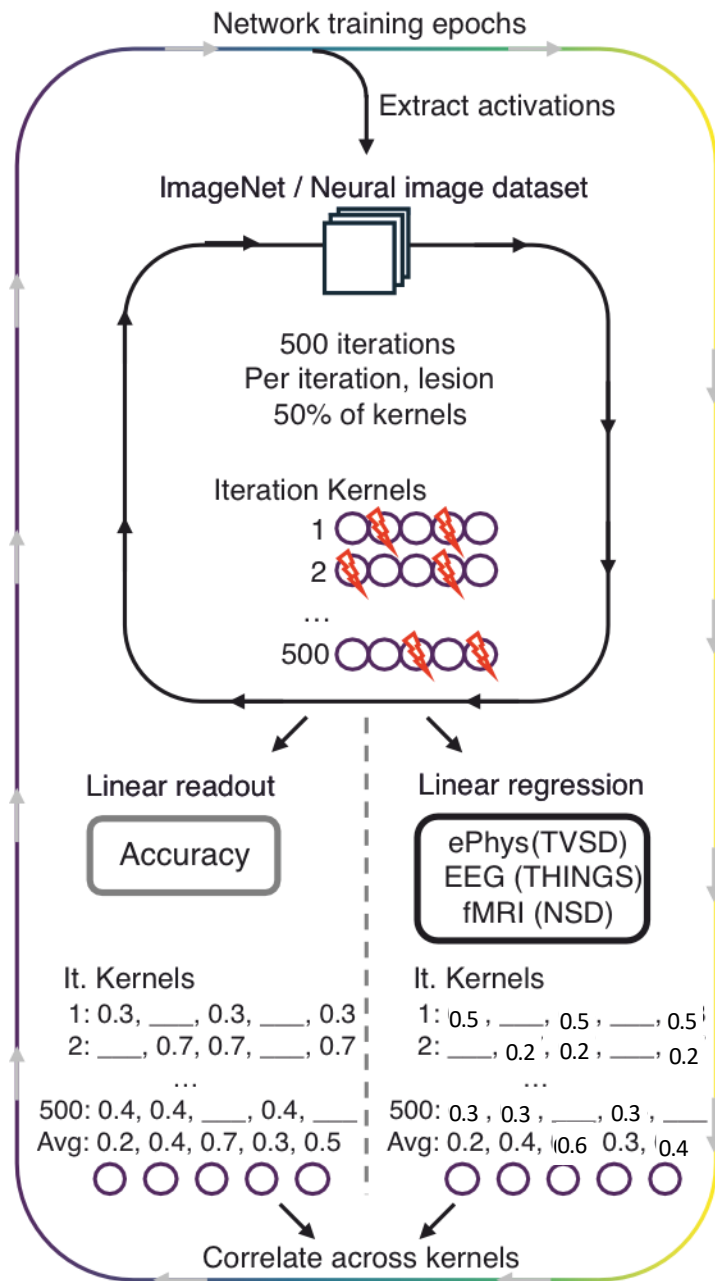
Epoch 38

Epoch 90
(fully trained)

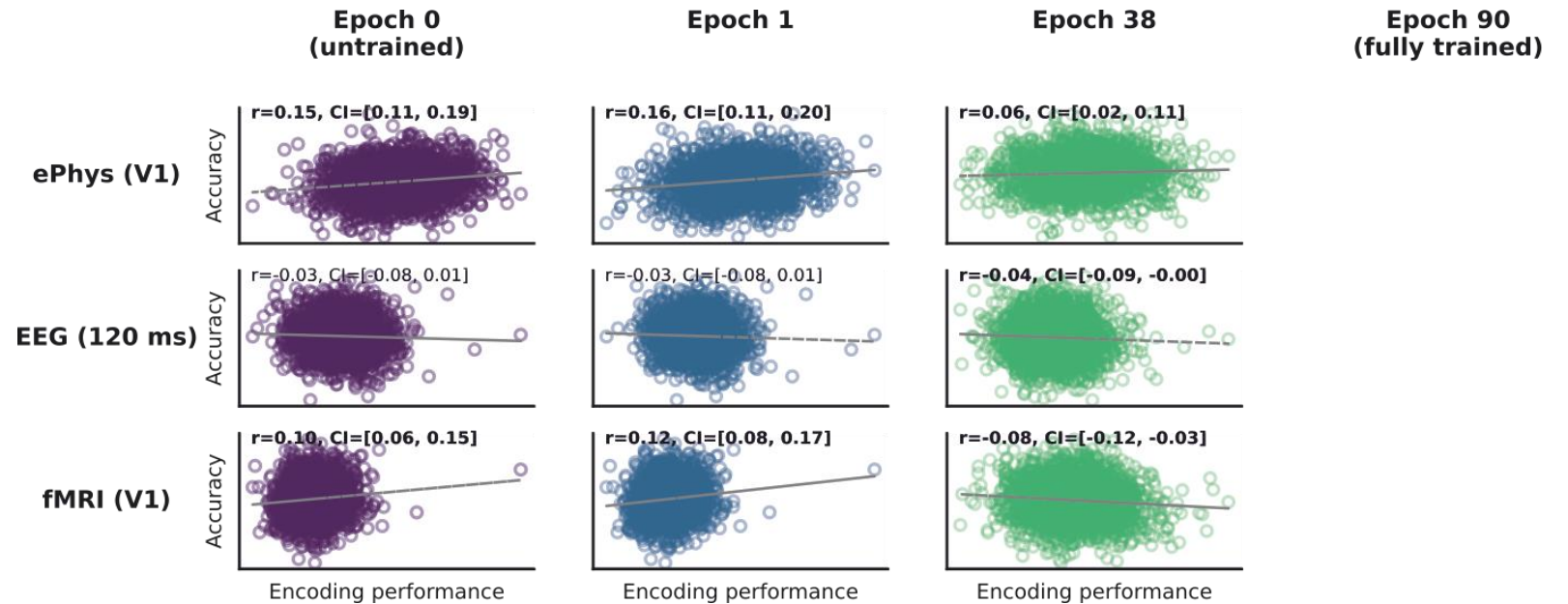
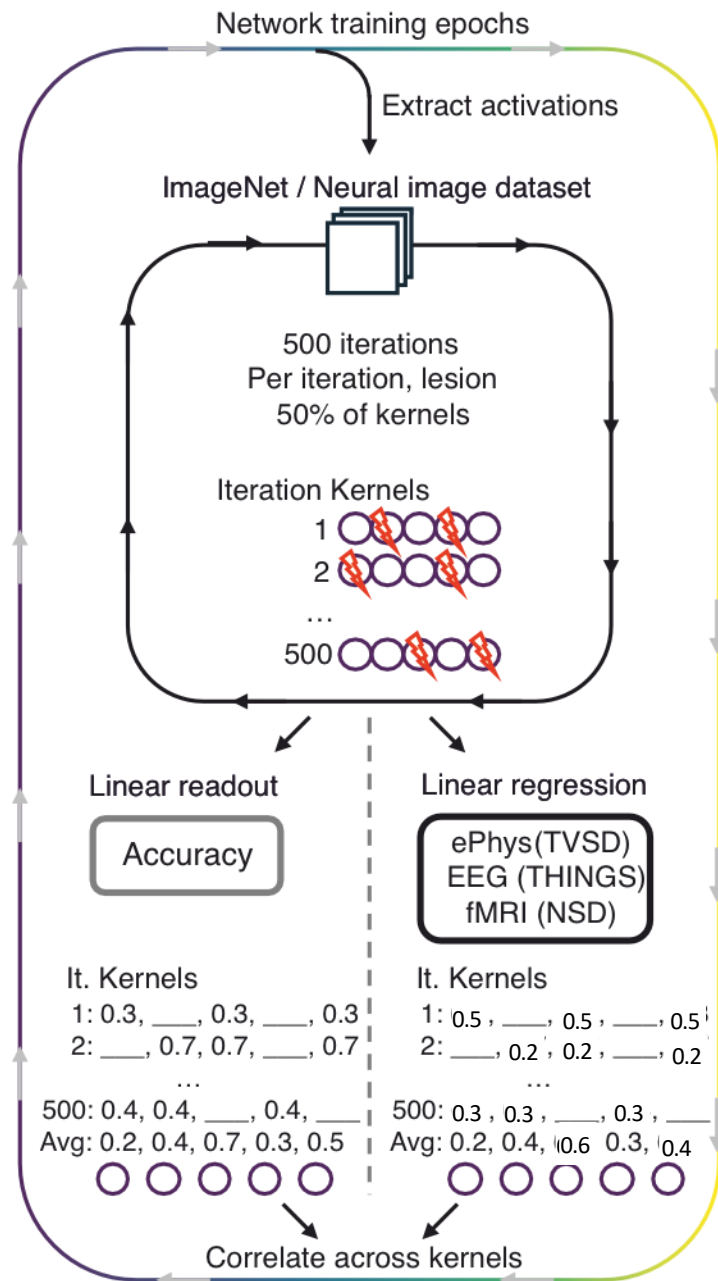
Kernel lesioning



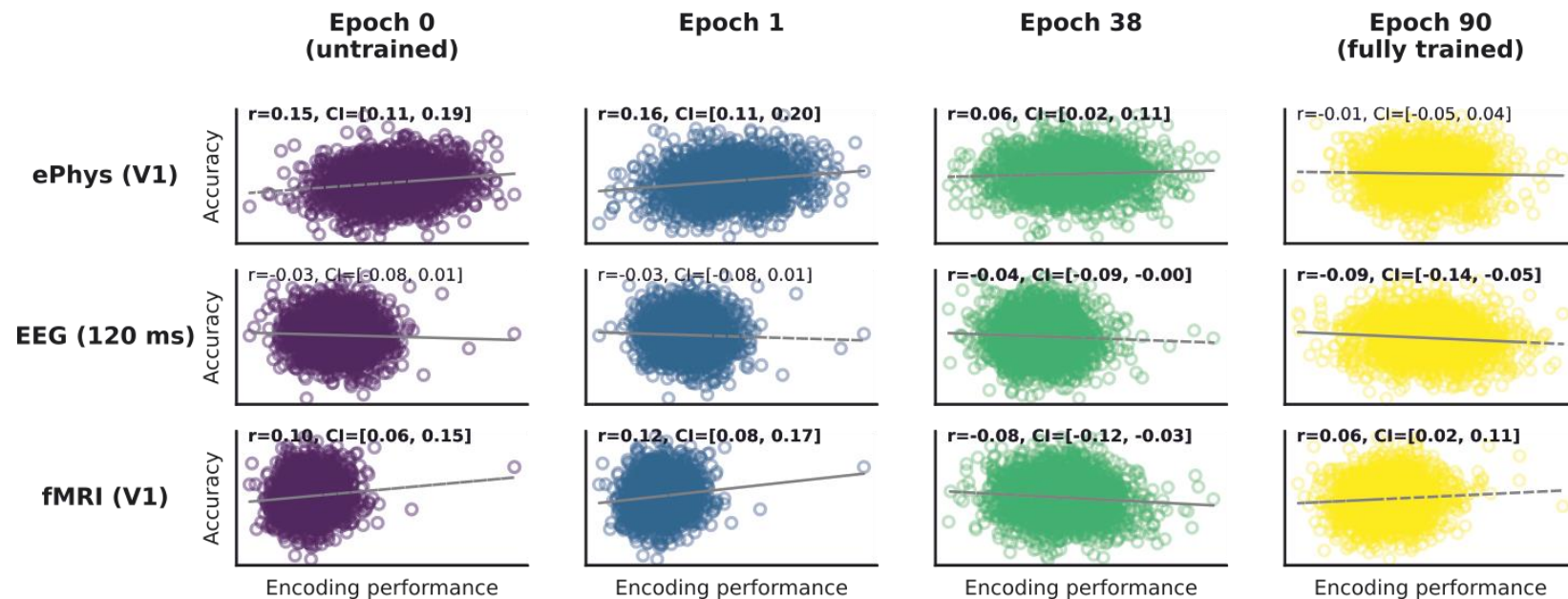
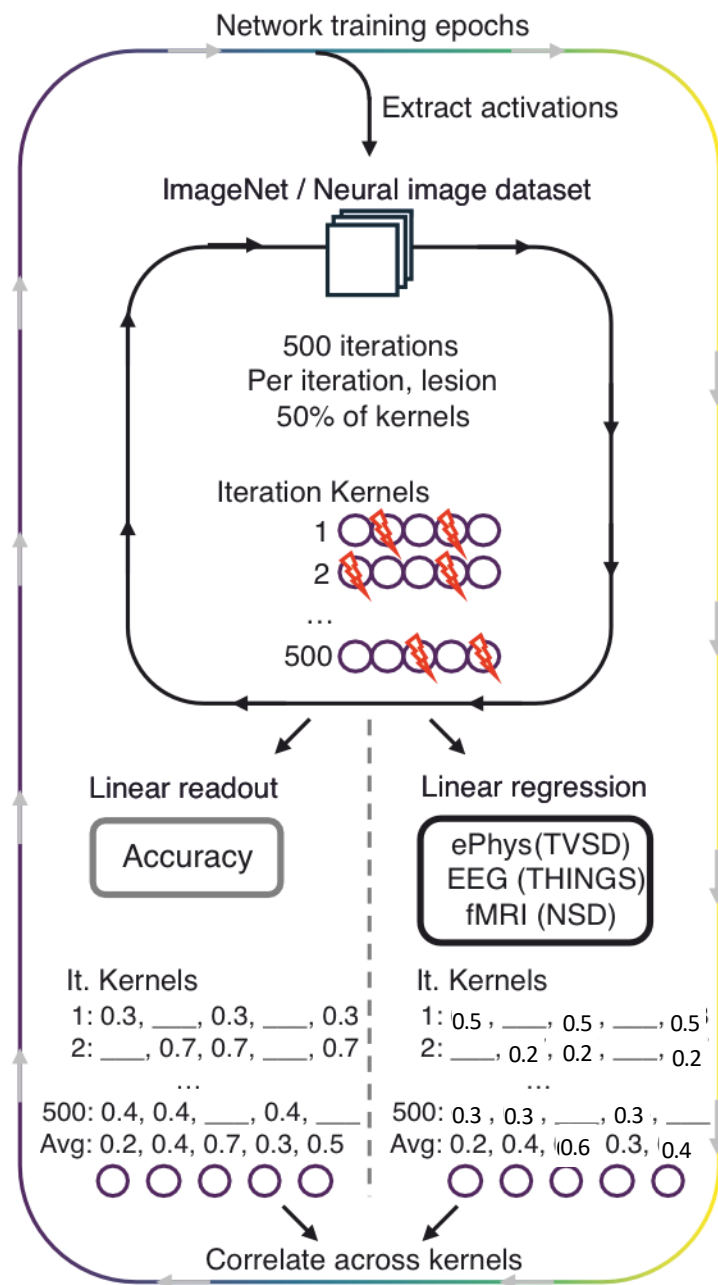
Kernel lesioning

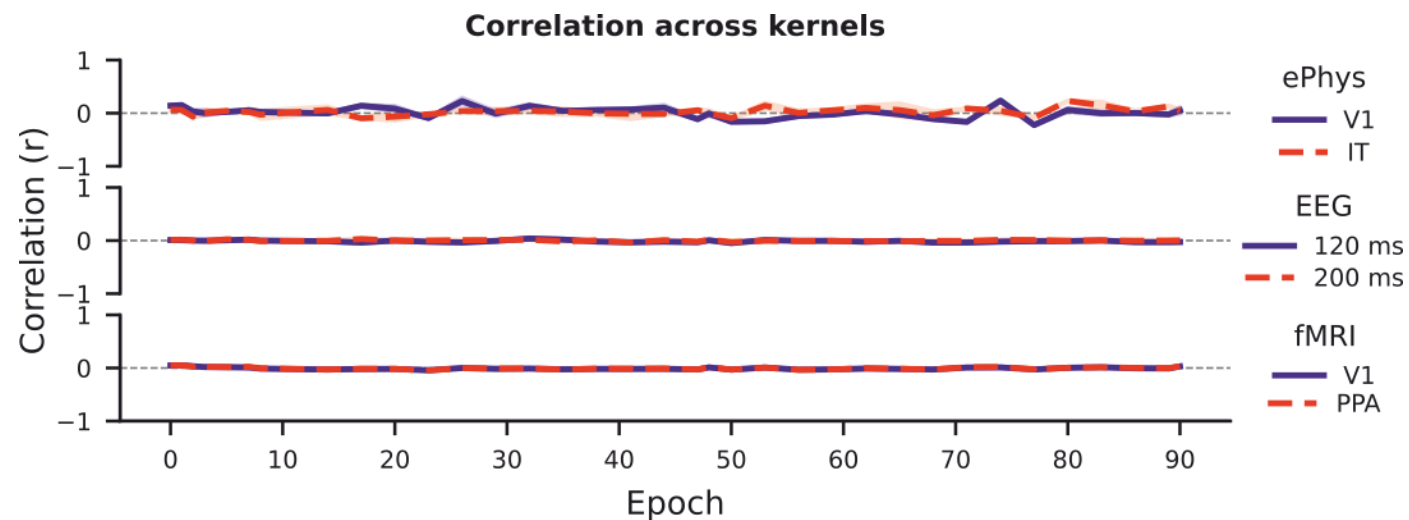
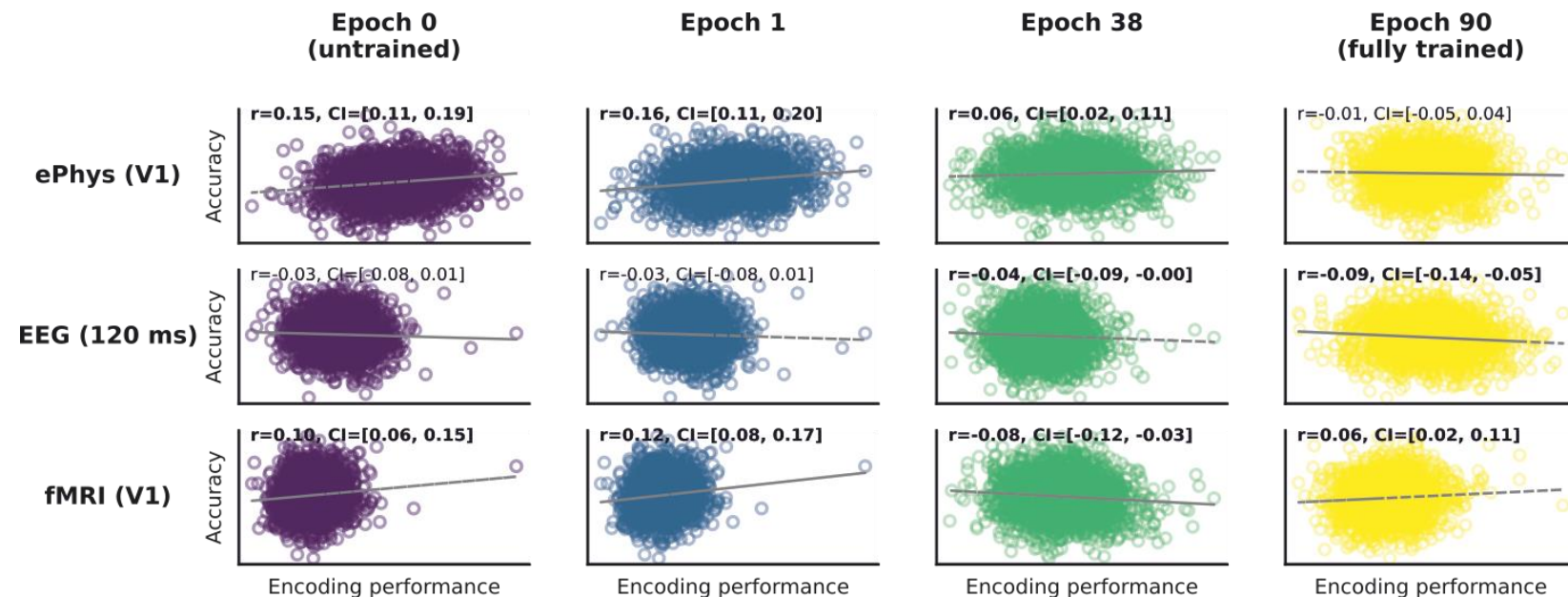
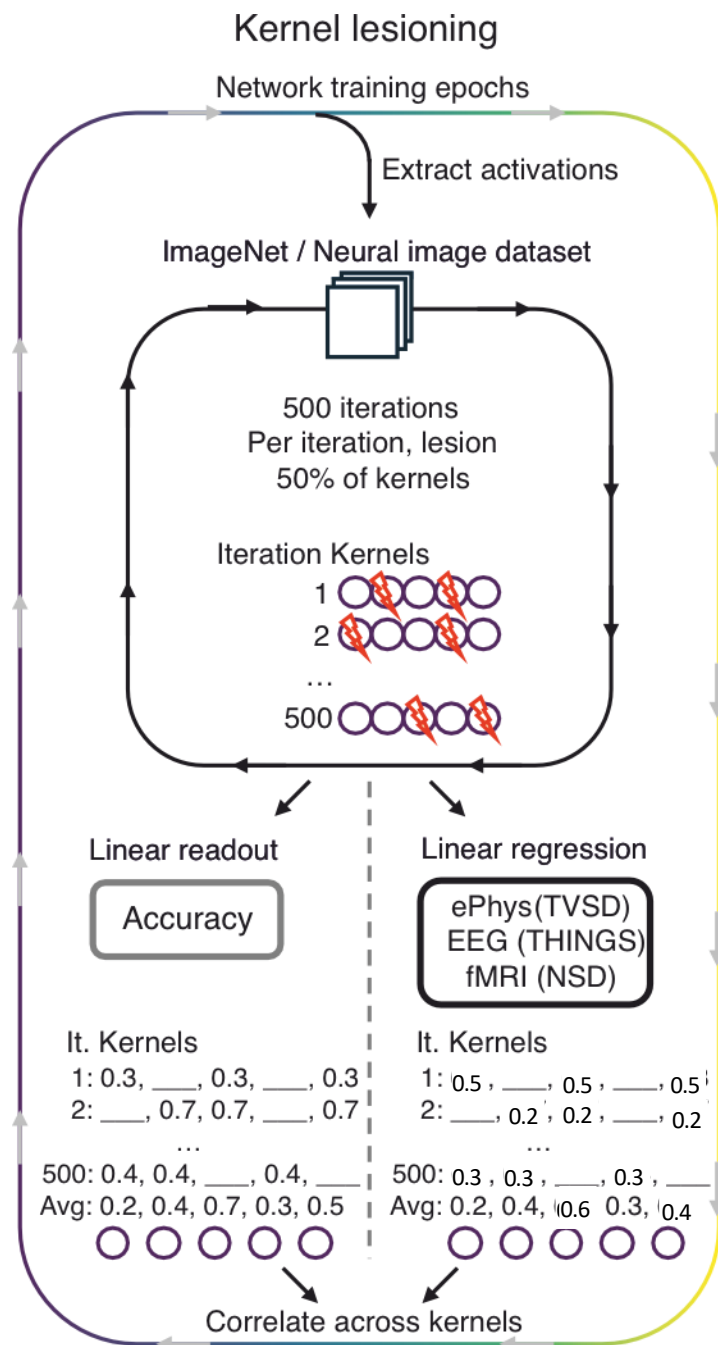


Kernel lesioning



Kernel lesioning





A shared visual world, not a shared solution to object recognition.

DCNN development



Repeating the
same task over
and over

Content: what is shared survives texture scrambling, so not global form.

Trajectory: brain alignment is established before classification exists

Features: alignment and classification run on largely separate kernels

This pattern of results is not consistent with a shared solution to object recognition.

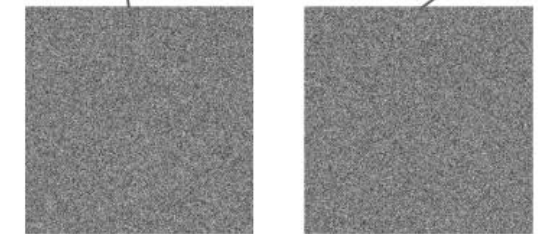
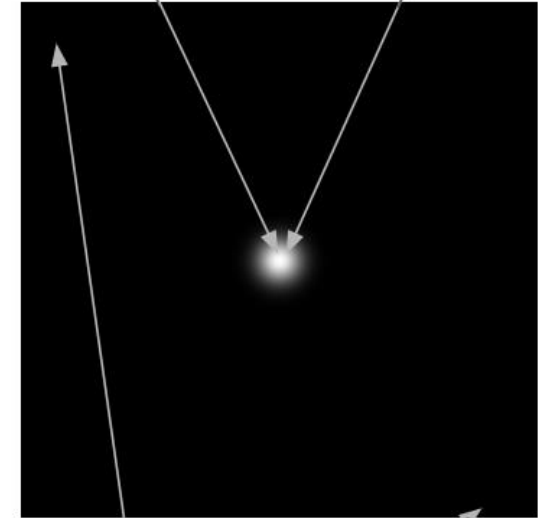
But it is consistent with that both are raised in a shared visual world.

Interactions with
many 'tasks' in a
curated world

Human development

Perceptual universe

$R = 0.7$

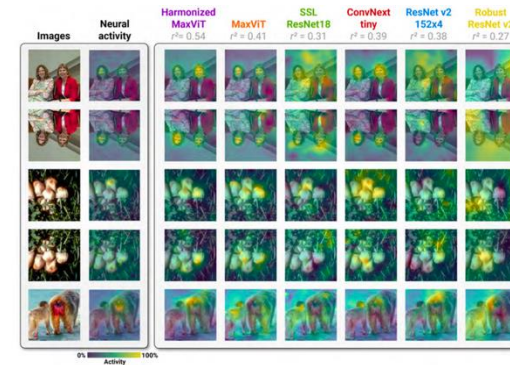


$R = 0$

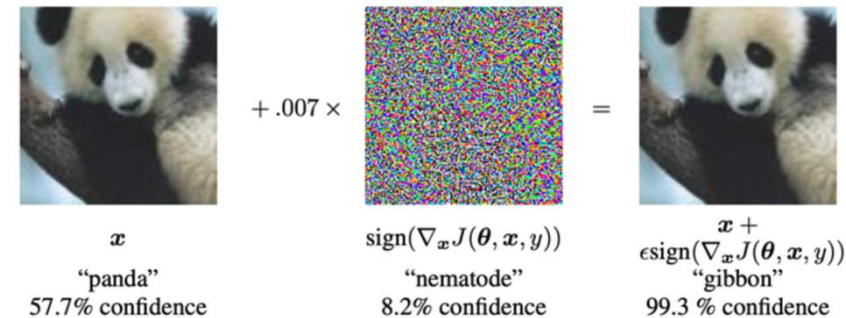
This view explains some of the problems associated with DCNNs

Highly trained networks will only move DCNNs further away from overlap.

DCNNs do not recognise objects the way humans do (but live in the same world).

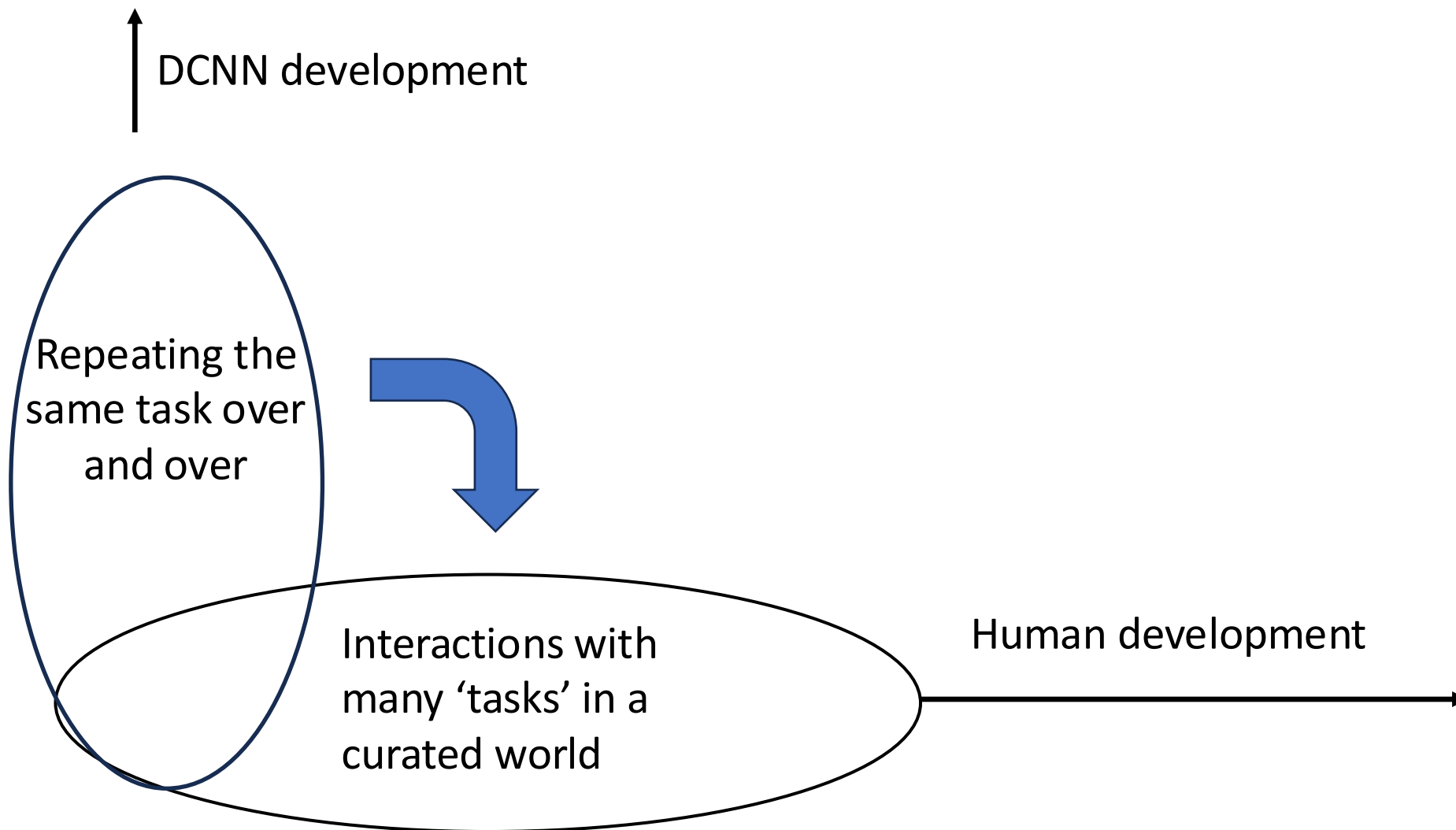


Linsley et al., 2023



Goodfellow et al., 2015

Improving the match between deep learning models and humans



Develop a nomenclature, not a labelling: terms that carve the behaviour at its joints and can be reasoned with

mind-sized, lossy.

extract

intervene

Model

world-sized, opaque, predicts

Seems sensible to divide DCNN – brain alignment in 3 stages

- **Initial alignment:**

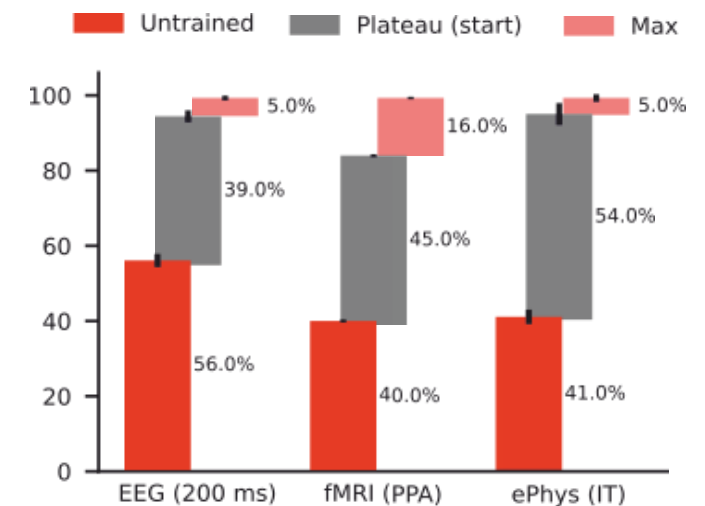
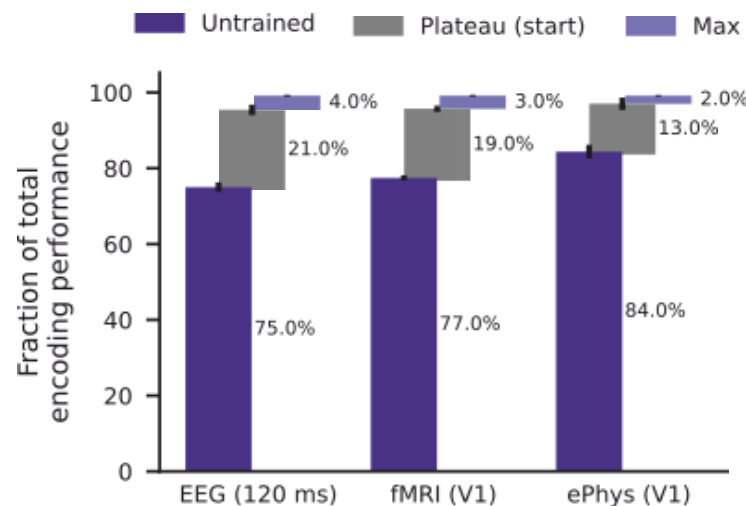
Cichy et al 2016. Bonner et al. 2026

- **Rapid increase**

~first 3 epochs

- **Plateau**

Changes within the SEM



Develop a nomenclature, not a labelling: terms that carve the behaviour at its joints and can be reasoned with

mind-sized, lossy.

↑ DCNN development

extract

intervene

Model

world-sized, opaque, predicts

Repeating the
same task over
and over

Interactions with
many 'tasks' in a
curated world

Human development



Theories, what are they good for

Two paradigms of perception

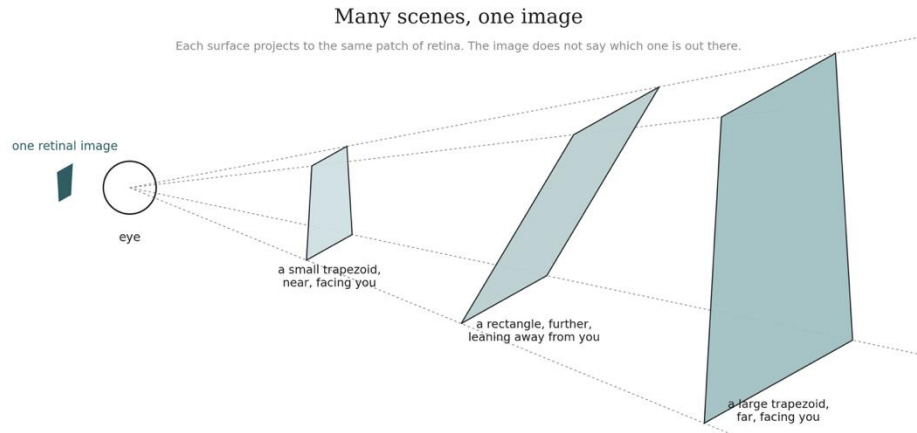
Mind-sized

Two levels

A shared world, not a shared objective

Empirical theory : computational cognitive neuroscience

The two paradigms, recapitulated



Many scenes, one image. The image alone does not fix the scene. Both paradigms start here.

The observer settles it (Helmholtz).

Reconstruct the scene from features; act on the reconstruction.

Its problem: binding.

The world settles it (Lorenz).

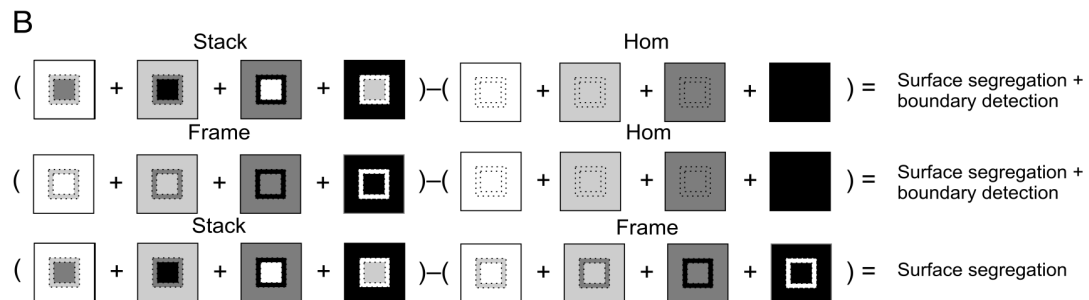
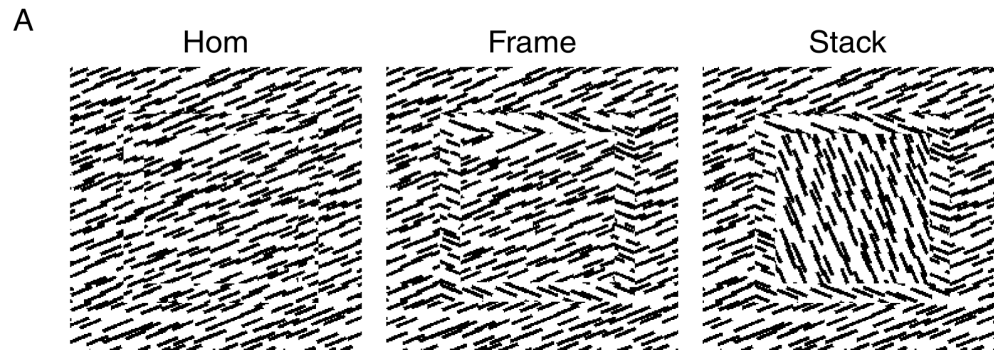
The scenes are not equally likely; act on the regularities without reconstructing.

Its problem: reach.

Representation, composition, and the role of attention

Helmholtz (cognitive psychology)		Lorenz (NeuroConnectionism)
Dimension		
Representation	Discrete symbols in separate data structures.	Deep hierarchical, mixed-selectivity population codes.
Localisation of information	Discrete object tokens tied to spatial pointers.	Hierarchical subspaces with continuity; sparsity and expressiveness increase with depth.
Composition (how objects emerge)	JOIN-Like binding across segregated maps.	Simple readout over population geometry; many conjunctions are linearly decodable from intermediate activity.
Role of attention/control	Driver of construction (creates object tokens).	Attention/control: reweights/routes info as needed.
Temporal signature	Late, attention-timed binding stage.	Fast feedforward for typical input; selective recurrent/attention for atypical cases or tasks.
Development	A miracle happens	Common co-occurrences densify and rare once prune. Practice on rare pairs improves local separability.
Failure modes	Sensitive to noise; heavy combinatorial bookkeeping	Struggles with deviations from learned structure; than leans on attention/control
Take-home	Binding = JOIN on separate tables	Readout over geometry replaces bind(); attention/control handles exceptions

Very much an example of the
 helmholtz branch research. Looking
 for a **function** (segmentation) and
 using **idealised stimuli** & a **mind sized**
framework

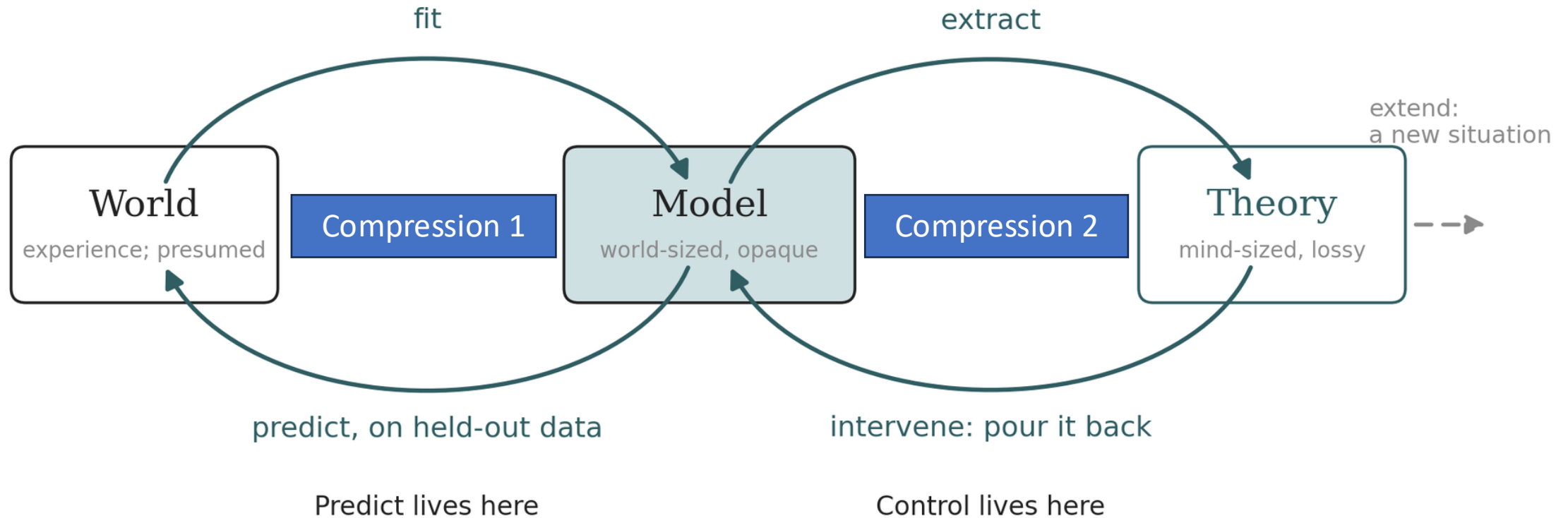


Very much an example of the lorenze
 branch research. Using as much
feature co-structure from the world
 as possible.



What empirical theory could mean

A mind-sized nomenclature, extracted from systems that work, checked in them, judged by its.



Lives in a broader 'paradigm' (meta-science) in which specific questions are asked, of certain data, looking for specific answers



Noor Seijdel



Niklas Müller



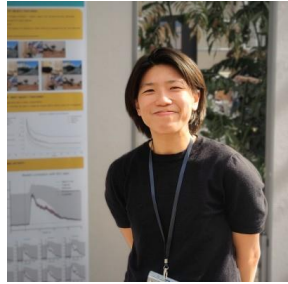
Julio Smidi



Denny Borsboom



Marcel van Gerven



Jessica Loke



Lynn Sörenson



Jolien Francken



Iris Groen

Thank you for your attention

When does alignment occur?

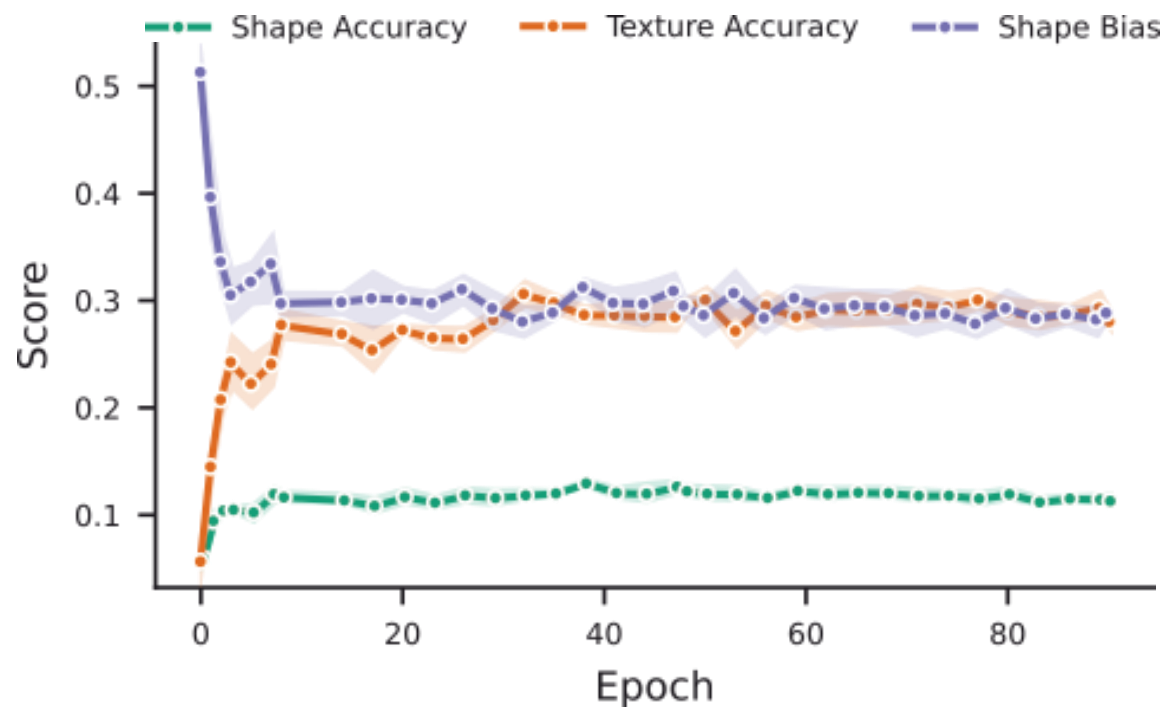
texture



shape



Challenge image



Shape accuracy arrives in the initial phase.

Texture accuracy keeps on increasing substantially longer.

This results in a drop of shape bias.

Suggest texture bias is not a property of the network; it is a product of late training.